



BANARAS HINDU UNIVERSITY

B.Sc.(Hons.) Courses

Effective from 2024 under NEP

Offered by

Faculty of Science



BANARAS HINDU UNIVERSITY



Department of Zoology

Faculty of Science

SYLLABI

B.Sc. (Hons.) Programme

In

Zoology

NEP COURSE STRUCTURE FOR 4-YEAR UG ZOOLOGY
Department of Zoology, Banaras Hindu University

Semester –I		
Course Code	Title	Credits
ZOOMJ11	Animal Diversity and Animal Form & Function	3
ZOOMJ12	Lab. exercises based on course ZOOMJ11	1
ZOOMN11	Animal Diversity and Animal Form & Function	3
ZOOMN12	Lab. exercises based on course ZOOMN11	1
ZOOMD11	Animal Biology	3
ZOOSE11	Applied Zoology	2
ZOOSE12	Lab. exercises based on course ZOOSE11	1
AE11	Offered by Institute/University	2
VA11	Offered by Institute/University	2
VA12	Offered by Institute/University	2
Total		20
Semester –II		
ZOOMJ21	Fundamentals of Cell Biology and Biochemistry	3
ZOOMJ22	Lab. exercises based on course ZOOMJ21	1
ZOOMN21	Fundamentals of Cell Biology and Elementary Biochemistry	3
ZOOMN22	Lab. Exercises based on course ZOOMN21	1
ZOOMD21	Hormones and Diseases	3
ZOOSE21	Food, Preservatives and Adulterants	2
ZOOSE22	Lab. exercises based on course ZOOSE21	1
AE21	Offered by Institute/University	2
VA21	Offered by Institute/University	2
VA22	Offered by Institute/University	2
Total		20
Semester –III		
ZOOMJ31	Basic Genetics	3
ZOOMJ32	Economic Zoology	3
ZOOMJ33	Lab. exercises based on course ZOOMJ31 & ZOOMJ32	2
ZOOMV31	Economic Zoology	3
ZOOMV32	Lab. exercises based on course ZOOMV31	1
ZOOMD31	Archaeozoology	3
ZOOSE31	Microscopy and Histology & Histochemistry	2
ZOOSE32	Lab. exercises based on course ZOOSE31	1
AE31	Offered by Institute/University	2
Total		20

Semester –IV		
ZOOMJ41	Fundamental Endocrinology	3
ZOOMJ42	Developmental Biology	3
ZOOMJ43	Evolution	2
ZOOMJ44	Systematics and Taxonomy	2
ZOOMJ45	Lab. exercises based on course ZOOMJ41 & ZOOMJ42	2
ZOOMJ46	Lab. exercises based on course ZOOMJ43 & ZOOMJ44	2
ZOOMN41	Fundamental Endocrinology	3
ZOOMN42	Lab. exercises based on course ZOOMN41	1
AE41	Offered by Institute/University	2
Total		20
Semester –V		
ZOOMJ51	Functional Anatomy of Non-chordates	3
ZOOMJ52	Animal Behaviour	2
ZOOMJ53	Immunology	3
ZOOMJ54	Biology of Infectious Diseases	2
ZOOMJ55	Lab. exercises based on course ZOOMJ51 & ZOOMJ52	2
ZOOMJ56	Lab. exercises based on course ZOOMJ53 & ZOOMJ54	2
ZOON57	Internship	2
ZOOMN51	Aquaculture	3
ZOOMN52	Lab. exercises based on course ZOOMN51	1
Total		20
Semester –VI		
ZOOMJ61	Functional Anatomy of Chordates	3
ZOOMJ62	Environmental Biology	3
ZOOMJ63	Cell Biology	3
ZOOMJ64	Mammalian Physiology	3
ZOOMJ65	Lab. exercises based on course ZOOMJ61 & ZOOMJ62	2
ZOOMJ66	Lab. exercises based on course ZOOMJ63 & ZOOMJ64	2
ZOOMV61	Clinical Biochemistry	3
ZOOMV62	Lab. exercises based on course ZOOMV61	1
Total		20

Semester –VII		
ZOOMJ71	Biochemistry and Molecular Biology	3
ZOOMJ72	Molecular Genetics	3
ZOOMJ73	Molecular Endocrinology	3
ZOOMJ74	Bioinformatics and Biostatistics	3
ZOOMJ75	Lab. exercises based on course ZOOMJ71 & ZOOMJ72	2
ZOOMJ76	Lab. exercises based on course ZOOMJ73 & ZOOMJ74	2
ZOOMN71	Structure and Functions of Biomolecules	3
ZOOMN72	Lab. exercises based on course ZOOMN71	1
Total		20
Semester –VIII (Honours)		
ZOOMJ81	Biotechniques	3
ZOOMJ82	Neurobiology	3
ZOOMJ83	Chronobiology	3
ZOOMJ84	Entomology and Fish Biology	3
ZOOMJ85	Lab. exercises based on course ZOOMJ81	1
ZOOMJ86	Lab. exercises based on course ZOOMJ82 & ZOOMJ83	2
ZOOMJ87	Lab. exercises based on course ZOOMJ84	1
ZOOMN81	Biotechniques	3
ZOOMN82	Lab. exercises based on course ZOOMN81	1
Total		20
Semester –VIII (Honours with Research)		
ZOOMJ8R1	Biotechniques	3
ZOOMJ8R2	Lab. exercises based on course ZOOMJ8R1	1
ZOOMN8R1	Biotechniques	3
ZOOMN8R2	Lab. exercises based on course ZOOMN8R1	1
ZOODS8R3	Dissertation	12
Total		20

Note: Students may have the option to take Offline as well as Online courses. However, students may take up to the 40% of the credits from Online courses in each category. Further, the capping of 40% may be relaxed to 60% for Courses in "Others" category to enable them to credit more interesting courses. Students will have to pay the fees for Online courses.

MJ – Major Course; MN – Minor Course; MV – Minor (Vocational); MD – Multidisciplinary Course; SE – Skill Enhancement Course; AE – Ability Enhancement Course; VA – Value Added Course; IN – Internship; DS – Dissertation

Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

BANARAS HINDU UNIVERSITY
(Detailed Syllabus of different Courses under NEP 2020)

Department of Zoology
Faculty of Science

Course Title	ZOOMJ11: Animal Diversity and Animal Form & Function			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- Explore the origin and evolutionary connections between various phyla. - Examine the general characteristics of animals from protozoa to chordates. - Determine the taxonomic positions of hemichordates and protochordates. - Compare the structures and functions of animal physiological systems. - Investigate the adaptive modifications of animals in relation to their environments.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Criteria for classification of multicellular animals 1.1 Symmetry 1.2 Early development: Spiral and radial cleavage Origin of germ layers Protostomes (protostome subgroups) and Deuterostomes 1.3 Body cavities: acoelomates, pseudocoelomates, coelomates (schizo- and enterocoelomates) 1.4 Segmentation, metamerism and tagmatization 1.5 Homology and Analogy			3
II	2. Non-Chordates: General characters and classification of the following up to classes with examples showing distinctive features 2.1 Protozoa 2.2 Porifera 2.3 Cnidaria 2.4 Ctenophora 2.5 Platyhelminthes 2.6 Nematoda 2.7 Annelida 2.8 Arthropoda 2.9 Mollusca 2.10 Echinodermata			8
III	3. Hemichordates: General characters and classification up to sub-classes 4. Chordates: General characters and classification of the following up to sub-classes/ orders with examples showing distinctive/adaptive features 4.1 Protochordates: Urochordates, Cephalochordates 4.2 Cyclostomes 4.3 Pisces 4.4 Amphibians 4.5 Reptiles 4.6 Aves 4.7 Mammals			11
IV	5. Modes and Types of Feeding, Digestion and Excretion 5.1 Feeding mechanisms: suspension, deposit, cropping and sucking (herbivorous) and raptorial (carnivorous) 5.2 Intracellular and extracellular digestion: food vacuole and gastrovascular cavity 5.3 Open tubular: Metanephridia 5.4 Closed saccular: Malpighian tubules and vertebrate kidney			12

	6. Respiratory Organs and Pattern of Circulation 6.1 Structure and function of gills, trachea and vertebrate lungs 6.2 Pattern of circulation: Open and closed systems	
V	7. Nervous system, Receptors and Sense organ 7.1 Patterns of the nervous system in non-chordates 7.2 Organization of the nervous system in vertebrates 7.3 Phonoreception in mammals 7.4 Photoreception in mammals 8. Reproduction 8.1 Types of asexual reproduction: fission, regeneration and parthenogenesis 8.2 Sexual reproduction: primary sex organs and their functions	11
Texts / References	1. Hickman, Keen, Larson and Eisenhour: Animal Diversity (9 th ed., 2021, Mc Graw Hill) 2. Miller & Harley: Zoology (10 th ed. 2016, McGraw Hill) 3. Campbell & Reece: Biology (7th ed. 2005, Pearson) 4. Mader: Biology (9th ed. 2007, Brown) 5. Kotpal: Modern textbook of Zoology: Invertebrates (11 th ed. 2016 Rastogi) 6. Kotpal: Modern textbook of Zoology: Vertebrates (4 th ed. 2016 Rastogi)	
Learning Outcomes (not more than 100 words)	1. Students will develop a fundamental understanding of animal kingdom classification. 2. The ability to identify general characteristics of various phyla and classes, along with suitable examples in relation to phylogeny, will be acquired by students. 3. Understanding of the specialized structures and functions of various animal body parts will be gained. 4. Students will explore comparative and evolutionary perspectives of animal physiology.	

Course Title	ZOOMJ12: Lab. exercises based on course ZOOMJ11			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• To identify various phylum of animal kingdom based on their external features. • To study the adaptive features found in animals of various phyla. • To recognize morphological features and locate various organ systems in different animals. • To learn basic dissection skills required for anatomical understanding.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	<i>Part A: Animal Diversity</i> 1. Study of transverse sections/chart of the following: Sycon (as an example of Parazoa to show its structure, spicules and canal system), Hydra (as an example of diploblastic animal), Fasciola (as an example of a triploblastic acoelomate animal), Ascaris (as an example of triploblastic pseudocoelomate animal), Hirudinaria (as an example of triploblastic schizocoelomate animal), Frog (as an example of triploblastic enterocoelomate animal). 2. Study of salient features and classification up to classes of the following non-chordates with special emphasis on their adaptive characters: Amoeba, Euglena, Plasmodium, Paramecium, Euplectella, Physalia, Corallium, Hormiphora, Taenia, Ascaris (male and female), Nereis (including heteronereid stage), Hirudinaria, Bonellia, Chiton, Mytilus, Octopus, Limulus, Eupagurus, Sacculina, Asterias, Echinus, Holothuria.			8
II	3. Mounting of Obelia colony 4. Salient features and classification up to Orders of the following with special emphasis on their adaptive characters: Balanoglossus, Herdmania, Amphioxus Lamprey, Trygon, Lung Fish, Uraeotyphlus, Alytes, Hyla, Chameleon, Tortoise, poisonous and non-poisonous snakes, Duck, Kiwi, Duck-billed Platypus, Macropus, Mole, Lemur			7
III	<i>Part B: Animal Form & Function</i> 5.1. Dissection and mounting of cephalic appendages and Hastate plate of Palaemon 5.2. Dissection of the digestive system of Palaemon or Periplaneta americana 5.3. Dissection of afferent branchial arteries of Mystus			8
IV	6.1 Dissection of the nervous system of Palaemon 6.2 Permanent slide preparation of gemmules of sponges 6.3 Study of the following through permanent slides/museum specimens: Conjugation in Paramecium, Trochophore larva, Zoea larvae, Pluteus larvae, Tornaria, Ammocoetes and Tadpole (frog); Axolotl			7
Texts / References	1. Lal: Practical Zoology- Invertebrates (12 th ed., Rastogi Publications) 2. Lal: Practical Zoology- Vertebrates (12 th ed., Rastogi Publications) 3. Verma: A manual of Practical Zoology - Chordates (11 th ed., S Chand publications)			
Learning Outcomes (not more than 100 words)	1. Students will be able to identify animals belonging to various phyla based on their external morphology and adaptive features. 2. Students will be able to understand the structural organization and anatomical variability of different organ systems.			

Course Title	ZOOMN11: Animal Diversity and Animal Form & Function			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- Explore the origin and evolutionary relationships between different phyla. - Examine the general characteristics of animals from protozoa to chordates. - Analyze the taxonomic position of hemichordates and protochordates. - Compare the structures and functions of animal physiological systems. - Investigate the adaptive modifications in different animals in relation to their environment.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Criteria for classification of multicellular animals 1.1 Symmetry 1.2 Early development: Spiral and radial cleavage Origin of germ layers Protostomes (protostome subgroups) and Deuterostomes 1.3 Body cavities: acoelomates, pseudocoelomates, coelomates (schizo- and enterocoelomates) 1.4 Segmentation, metamerism and tagmatization 1.5 Homology and Analogy			3
II	2. Non-Chordates: General characters and classification of the following up to classes with examples showing distinctive features 2.1 Protozoa 2.2 Porifera 2.3 Cnidaria 2.4 Ctenophora 2.5 Platyhelminthes 2.6 Nematoda 2.7 Annelida 2.8 Arthropoda 2.9 Mollusca 2.10 Echinodermata			8
III	3. Hemichordates: General characters and classification up to sub-class 4. Chordates: General characters and classification of the following up to sub-classes/orders with examples showing distinctive/adaptive features 4.1 Protochordates: Urochordates, Cephalochordates 4.2 Cyclostomes 4.3 Pisces 4.4 Amphibians 4.5 Reptiles 4.6 Aves 4.7 Mammals			11
IV	5. Modes and Types of Feeding, Digestion and Excretion 5.1 Feeding mechanisms: suspension, deposit, cropping and sucking (herbivorous) and raptorial (carnivorous) 5.2 Intracellular and extracellular digestion: food vacuole and gastrovascular cavity 5.3 Open tubular: Metanephridia 5.4 Closed saccular: Malpighian tubules and vertebrate kidney 6. Respiratory Organs and Pattern of Circulation 6.1 Structure and function of gills, trachea and vertebrate lungs 6.2 Pattern of circulation: Open and closed systems			12
V	7. Nervous system, Receptors and Sense organ 7.1 Patterns of the nervous system in non-chordates 7.2 Organization of the nervous system in vertebrates 7.3 Phonoreception in mammals			11

	7.4 Photoreception in mammals 8. Reproduction 8.1 Types of asexual reproduction: fission, regeneration and parthenogenesis 8.2 Sexual reproduction: primary sex organs and their functions	
Texts / References	1. Hickman, Keen, Larson and Eisenhour: Animal Diversity(9th ed., 2021, Mc Graw Hill) 2. Miller & Harley: Zoology (10th ed. 2016, McGraw Hill) 3. Campbell & Reece: Biology (7th ed. 2005, Pearson) 4. Mader: Biology (9th ed. 2007, Brown) 5. Kotpal: Modern textbook of Zoology: Invertebrates (11th ed. 2016 Rastogi) 6. Kotpal: Modern textbook of Zoology: Vertebrates (4th ed. 2016 Rastogi)	
Learning Outcomes (not more than 100 words)	1. Students will gain a fundamental understanding of animal kingdom classification. 2. They will be able to identify the general characteristics of various phyla and classes, with suitable examples related to phylogeny. 3. An understanding of the specialized structures and functions of various animal body parts will be acquired. 4. Students will develop insights into the comparative and evolutionary aspects of animal physiology.	

Course Title	ZOOMN12: Lab. exercises based on course ZOOMN11			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• To identify various phylum of animal kingdom based on their external features. • To study the adaptive features found in animals of various phyla. • To recognize morphological features and locate various organ systems in different animals. • To learn basic dissection skills required for anatomical understanding.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Part A: Animal Diversity 1. Study of transverse sections/chart of the following: <i>Sycon</i> (as an example of Parazoa to show its structure, spicules and canal system), <i>Hydra</i> (as an example of diploblastic animal), <i>Fasciola</i> (as an example of a triploblastic acoelomate animal), <i>Ascaris</i> (as an example of triploblastic pseudocoelomate animal), <i>Hirudinaria</i> (as an example of triploblastic schizocoelomate animal), Frog (as an example of triploblastic enterocoelomate animal). 2. Study of salient features and classification up to classes of the following non-chordates with special emphasis on their adaptive characters: <i>Amoeba</i> , <i>Euglena</i> , <i>Plasmodium</i> , <i>Paramecium</i> , <i>Euplectella</i> , <i>Physalia</i> , <i>Corallium</i> , <i>Hormiphora</i> , <i>Taenia</i> , <i>Ascaris</i> (male and female), <i>Nereis</i> (including heteronereid stage), <i>Hirudinaria</i> , <i>Bonellia</i> , <i>Chiton</i> , <i>Mytilus</i> , Octopus, <i>Limulus</i> , <i>Eupagurus</i> , <i>Sacculina</i> , <i>Asterias</i> , <i>Echinus</i> , <i>Holothuria</i> .			8
II	3. Mounting of <i>Obelia</i> colony 4. Salient features and classification up to Orders of the following with special emphasis on their adaptive characters: <i>Balanoglossus</i> , <i>Herdmania</i> , <i>Amphioxus</i> , <i>Lamprey</i> , <i>Trygon</i> , Lung Fish, <i>Uraeotyphlus</i> , <i>Alytes</i> , <i>Hyla</i> , <i>Chameleon</i> , Tortoise, poisonous and non-poisonous snakes, Duck, Kiwi, Duck-billed Platypus, <i>Macropus</i> , Mole, <i>Lemur</i>			7
III	Part B: Animal Form & Function 5.1. Dissection and mounting of cephalic appendages and Hastate plate of <i>Palaemon</i> 5.2. Dissection of the digestive system of <i>Palaemon</i> or <i>Periplaneta americana</i> 5.3. Dissection of afferent branchial arteries of <i>Mystus</i>			8
IV	6.1 Dissection of the nervous system of <i>Palaemon</i> 6.2 Permanent slide preparation of gemmules of sponges 6.3 Study of the following through permanent slides/museum specimens: Conjugation in <i>Paramecium</i> , Trochophore larva, Zoa larvae, Pluteus larvae, Tornaria, Ammocoetes and Tadpole (frog); Axolotl			7
Texts / References	1. Lal: Practical Zoology- Invertebrates (12 th ed., Rastogi Publications) 2. Lal: Practical Zoology- Vertebrates (12 th ed., Rastogi Publications) 3. Verma: A manual of Practical Zoology - Chordates (11 th ed., S Chand publications)			
Learning Outcomes (not more than 100 words)	1. Students will be able to identify animals belonging to various phyla based on their external morphology and adaptive features. 2. Students will be able to understand the structural organization and anatomical variability of different organ systems.			

Course Title	ZOOMD11: Animal Biology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- Investigate the origin and evolutionary relationships between different phyla. - Delve into the basic units of life, biomolecules, and their functional organization. - Examine the flow of genetic information, genes, and related diseases. - Analyze the structures and functions of animal physiological systems.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Properties and origin of life			3
	2. Diversity of evolution of animal systems			8
	2.1 Classification of major groups of living system			
	2.2 Concept and evidence of organic evolution			
	2.3 Theory of natural selection – industrial melanism			
II	3. Cell – the unit of life			8
	3.1 Organization of cell			
	3.2 Cell multiplication			
III	4. Biomolecules			8
	4.1 Sugars, fats and proteins			
	4.2 Enzymes - biological catalysts			
	4.3 Production of energy in biological system			
IV	5. Inheritance of biological characters			8
	5.1 Laws of Inheritance			
	5.2 Nature of hereditary material and flow of genetic information			
	5.3 Genes and diseases – sickle cell anemia, Down syndrome			
V	6. Physiology of			10
	6.1 Digestion			
	6.2 Breathing			
	6.3 Blood circulation			
	6.4 Urine formation			
	6.5 Endocrine glands			
Texts / References	1. Biology for class XI & XII NCERT Textbooks 2. Verma et al, Cell Biology, Genetics, Molecular Biology, Evolution and Ecology (2008, S Chand) 3. Cain et al, Discover Biology (4 th Edition, W. W. Norton & Company)			
Learning Outcomes (not more than 100 words)	- Students will understand the origin and evolutionary relationship between different phyla and the basic units of life, biomolecules, and their functional organization. - They will able to connect the information on inheritance, genes and diseases. - They will understand the structures and functions of animal physiological systems.			

Course Title	ZOOSE11: Applied Zoology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		2
	Hour of Teaching (Total)	30		30
Course Objectives (not more than 100 words)	• To learn the basic culture methods of various economically important animals. • To learn the concept of aquaculture, bee products, silk production and animal husbandry. • To gain knowledge in the field of pest management and transgenic animals.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Pisciculture 1.1 Aquaculture resources in India 1.2 Fin fish culture			2
II	2. Economically important insects 2.1 Sericulture 2.1.1 Types, Life cycle and Rearing of Silkworms 2.1.2 Harvesting and processing of cocoons 2.1.3 Diseases and their control 2.2 Apiculture 2.2.1 Species, Life cycle and organization of bee colony 2.2.2 Bee products and their uses 2.2.3 Diseases and their control 2.3 Lac culture 2.3.1 Lac insect, Life cycle and its host plants 2.3.2 Lac production and its uses 2.3.3 Diseases and their control			14
III	3. Pests and their management 3.1 Cultural methods 3.2 Chemical methods 3.3 Biological methods			4
IV	4. Animal husbandry and Transgenic animals 4.1. Dairy 4.1.1 Breeds of Bovine 4.1.2 Techniques of dairy management 4.1.3 Milk and milk products 4.2 Poultry 4.2.1 Types of Breeds 4.2.2 Reproductive system and egg formation (Hen model) 4.2.3 Nutritional value of egg and chicken 4.3 Transgenic animals			10
Texts / References	1. Shukla and Upadhyaya: Economic Zoology (Rastogi Publishers) 2. Omkar: Industrial Entomology (Springer Publisher, 2017) 3. Omkar: Pests and Their Management (Springer Publisher, 2018)			
Learning Outcomes (not more than 100 words)	1. Students will gain insight into aquaculture systems, including breeding and post-harvest techniques. 2. They will explore the concepts of beekeeping, silkworm rearing, and the associated pests and diseases. 3. Understanding of various types of animal husbandry and their management practices will be developed.			

Course Title	ZOOSE12: Lab. exercises based on course ZOOSE11			
Category of Course	Major / Minor / Minor (Vocational)/ SEC /AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• Learn about the features of various economically important animals. • Study the life cycle of economically significant insects and their applications. • Identify adulterations in milk and evaluate the quality of eggs.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Dairy 1.1 External morphology and nomenclature of dairy animals. 1.2 Different types of cattle sheds and schedule of day-to-day operation of the dairy farm. 1.3 Determination of the specific gravity of milk by using a mercury lactometer.			8
II	2. Poultry 2.1 External morphology of poultry birds (model) 2.2 Test for fertilized and unfertilized eggs (Light test, Cracking test). 2.3 Test for egg quality (Floating test, cracking test).			6
III	3. Fishery 3.1. External morphology of fish. 3.2 Identification of two major carps – <i>Labeo rohita</i> and <i>Catla catla</i> .			6
IV	4. Economically important insects 4.1 Honey Bee i. Castes and life cycle (model/chart). ii. Mounting of the sting apparatus. iii. Whole mount preparation of the 3 pairs of legs. 4.2 Silkworm i. Life cycle of mulberry silkworm, <i>Bombyx mori</i> ii. Life cycle of tasar silkworm, <i>Antheraea mylitta</i> .			10
Learning Outcomes (not more than 100 words)	1. Students will understand the external morphology and management of various poultry, dairy breeds and quality of eggs and milk. 2. Students will understand life cycle of economically important insects.			

Course Title	ZOOMJ21: Fundamental of Cell Biology and Biochemistry			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	<i>Fundamentals of Cell Biology:</i> - This core Zoology course provides a comprehensive understanding of eukaryotic animal cells, including their organelles and cytoskeleton. - It covers mitochondrial organization, cellular bioenergetics, and homeostasis, enabling students to grasp both simple and complex cellular processes. <i>Elementary Biochemistry:</i> - This course underpins advances in medicine, veterinary medicine, agriculture, and biotechnology. - It covers how cells operate through interconnected chemical reactions, the role of macromolecules (proteins, carbohydrates, lipids, and nucleic acids) in cellular functions, and applications in product development and healthcare advancements.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content	Hr. of Teaching		
I	1. The Cell 1.1 Pro and Eukaryotic Cell 1.2 Methods in Cell Biology: Idea of microscopy and cell fractionation 2. Organization of cell 2.1 Structure and function of plasma membrane: Lipid bilayer, Cortical cytoskeleton, Membrane proteins, Active and passive transport. 2.2 Introduction to endomembrane system 2.2.1 Endoplasmic reticulum: Forms and Co-translational transport, 2.2.2 Golgi complex: Protein sorting, retrograde pathway 2.2.3 Lysosome and peroxisome 2.3 Introduction to cytoskeleton: Microtubules, Microfilaments and Intermediate filaments 2.4 Mitochondria 2.4.1 Mitochondrial genome and Endo-symbiotic hypothesis 2.4.2 Structure and Functions: Membrane proteins, Chemiosmotic process 2.5 Nuclear envelope and nuclear pore complex 2.6 Nucleolus and Biogenesis of ribosome 2.7 Interphase chromatin, Chromosomes 2.8 Higher order chromatin organization, heterochromatin and euchromatin, chromosome bands 2.9 Polytene and lamp brush chromosomes as models of gene regulation	13		
II	3. Cell reproduction 3.1 Basic features of cell cycle 3.2 Cell Cycle checkpoints and their regulation (Restriction point, Cyclins, Cdks and APC) 3.3 Mitosis: Microtubule complex and spindle assembly 3.4 Process and phases of meiosis and its significance 4. Elementary knowledge of cell transformation and cancer 5. Introduction to the cellular basis of immunity	10		
III	Elementary Biochemistry Introduction: 1.1 Scope and Importance of Biochemistry 1.2 Principle of biomolecular organization, configuration and conformation 1.3 Water as a biological solvent	2		
IV	Proteins and Enzymes	9		

	2.1 Functions and Diversity of Proteins 2.2 Properties of peptide bond 2.3 Structural organization of proteins 2.4 General properties of enzymes 2.5 Major classes of enzymes 2.6 Mechanism of enzyme action (ES complex and lowering of activation energy and Concept of chemical catalysis) 2.7 Introduction to enzyme kinetics	
V	Carbohydrates and Lipids 3.1 Classification of carbohydrates 3.2 Structure and conformation of monosaccharides 3.3 Oligosaccharides (disaccharides) and polysaccharides 3.4 Biological significance and classification of lipids 3.5 Fatty acids: Types and nomenclature Nucleic acids 4.1 Bases, nucleosides and nucleotides 4.2 DNA structure: DNA double helix (Watson and Crick model) 4.3 Basic mechanism of DNA replication (Prokaryotes) 4.4 Transcriptional unit and basic concept of transcription (Prokaryotes) 4.5 Genetic code and basic mechanism of translation (Prokaryotes) 4.6 Introduction to recombinant DNA techniques and their applications	11
Texts / References	<i>Fundamental of Cell Biology</i> 1. Alberts et al, Molecular Biology of the Cell (7 th Edition, Garland) 2. Karp: Cell and Molecular Biology (9 th Edition, John Wiley) 3. Lodish et al: Molecular Cell Biology (9 th Edition, Freeman) 4. Plopper et al, Lewin's Cells (3 rd Edition, Jones and Bartlett) <i>Elementary Biochemistry</i> 1. Berg, Tymoczko, Stryer; Biochemistry (7 th ed. 2012, Freeman) 2. Conn, Stumpf, Bruening; Principles of Biochemistry (5 th ed. 1987, Wiley)	
Learning Outcomes (not more than 100 words)	1. The students shall have a thorough understanding of the chemical components of the cell, their structure, and functions. 2. Gain insight into the chemical reactions and interactions between the macromolecules regulating cellular machinery. 3. Use the knowledge gained in making scientific breakthroughs in disease biology and transform global health care. 4. Useful for a career in biomedical research, forensic science, and pharmaceuticals.	

Course Title	ZOOMJ22: Lab. exercises based on course ZOOMJ21			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• The practical course aims to develop students' proficiency in basic laboratory techniques, scientific experimentation, hypothesis testing, and data interpretation while fostering good lab practices and independent problem-solving skills. • Students will master microscopy and staining techniques to analyze eukaryotic cells and their organelles. • Practical skills in cell biology will be developed through techniques like centrifugation, staining, and observing mitosis. • Students will perform biochemical tests to understand fundamental biochemical processes.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content-			Hr. of Teaching
I	<i>Part A: Fundamentals of Cell Biology</i> 1. Drawing of the ultrastructure of the cell and different organelles (from photographs provided). 2. Familiarization with the student's light microscope and stereobinocular microscope. 3. Application of centrifuge – separation of sperm from other testicular cells by low-speed centrifugation. 4. Diversity of eukaryotic cells – methylene blue staining of buccal epithelium, sperm, neurons, striated muscle cells; Leishman staining of mammalian blood cells. 5. Permeability of plasma membrane – effect of isotonic, hypotonic and hypertonic solutions on mammalian RBC. 6. Staining of mitochondria with Janus green in the buccal epithelium. 7. Mitosis in onion root tip.			15
II	<i>Part B: Elementary Biochemistry</i> 1. Preparation of models of amino acids, dipeptides and nucleotides. 2. Ninhydrin test for α-amino acids. 3. Benedict's test for reducing sugars. 4. Iodine test for starch. 5. Determination of acid value of oil.			15
Texts / References				
Learning Outcomes (not more than 100 words)	• The students must be able to perform experiments independently and become proficient in basic lab techniques. • Gain in-depth understanding of biochemical principles, data interpretations and logical conclusions.			

Course Title	ZOOMN21: Fundamental of Cell Biology & Elementary Biochemistry			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	3
	Hour of Teaching (Total)	45	30	45
Course Objectives (not more than 100 words)	<i>Fundamentals of Cell Biology:</i> - This core Zoology course provides a comprehensive understanding of eukaryotic animal cells, including their organelles and cytoskeleton. - It covers mitochondrial organization, cellular bioenergetics, and homeostasis, enabling students to grasp both simple and complex cellular processes. <i>Elementary Biochemistry:</i> - This course underpins advances in medicine, veterinary medicine, agriculture, and biotechnology. - It covers how cells operate through interconnected chemical reactions, the role of macromolecules (proteins, carbohydrates, lipids, and nucleic acids) in cellular functions, and applications in product development and healthcare advancements.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content-	Hr. of Teaching		
I	1. The Cell 1.1 Pro and Eukaryotic Cell 1.2 Methods in Cell Biology: Idea of microscopy and cell fractionation 2. Organization of cell 2.1 Structure and function of plasma membrane: Lipid bilayer, Cortical cytoskeleton, Membrane proteins, Active and passive transport. 2.2 Introduction to endomembrane system 2.2.1 Endoplasmic reticulum: Forms and Co-translational transport, 2.2.3 Golgi complex: Protein sorting, retrograde pathway 2.2.3 Lysosome and peroxisome 2.3 Introduction to cytoskeleton: Microtubules, Microfilaments and Intermediate filaments 2.4 Mitochondria 2.4.1 Mitochondrial genome and Endo-symbiotic hypothesis 2.4.2 Structure and Functions: Membrane proteins, Chemiosmotic process 2.5 Nuclear envelope and nuclear pore complex 2.6 Nucleolus and Biogenesis of ribosome 2.7 Interphase chromatin, Chromosomes 2.8 Higher order chromatin organization, heterochromatin and euchromatin, chromosome bands 2.9 Polytene and lamp brush chromosomes as models of gene regulation	13		
II	3. Cell reproduction 3.1 Basic features of cell cycle 3.2 Cell Cycle checkpoints and their regulation (Restriction point, Cyclins, Cdks and APC) 3.3 Mitosis: Microtubule complex and spindle assembly 2.4 Process and phases of meiosis and its significance 3. Elementary knowledge of cell transformation and cancer 4. Introduction to the cellular basis of immunity	10		
III	Elementary Biochemistry Introduction: 1.1 Scope and Importance of Biochemistry 1.2 Principle of biomolecular organization, configuration and conformation 1.3 Water as a biological solvent	2		
IV	Proteins and Enzymes 2.1 Functions and Diversity of Proteins	9		

	2.2 Properties of peptide bond 2.3 Structural organization of proteins 2.4 General properties of enzymes 2.5 Major classes of enzymes 2.6 Mechanism of enzyme action (ES complex and lowering of activation energy and Concept of chemical catalysis) 2.7 Introduction to enzyme kinetics	
V	Carbohydrates and Lipids 3.1 Classification of carbohydrates 3.2 Structure and conformation of monosaccharides 3.3 Oligosaccharides (disaccharides) and polysaccharides 3.4 Biological significance and classification of lipids 3.5 Fatty acids: Types and nomenclature Nucleic acids 4.1 Bases, nucleosides and nucleotides 4.2 DNA structure: DNA double helix (Watson and Crick model) 4.3 Basic mechanism of DNA replication (Prokaryotes) 4.4 Transcriptional unit and basic concept of transcription (Prokaryotes) 4.5 Genetic code and basic mechanism of translation (Prokaryotes) 4.6 Introduction to recombinant DNA techniques and their applications	13
Texts / References	<i>Fundamental of Cell Biology</i> 5. Alberts et al, Molecular Biology of the Cell (7 th Edition, Garland) 6. Karp: Cell and Molecular Biology (9 th Edition, John Wiley) 7. Lodish et al: Molecular Cell Biology (9 th Edition, Freeman) 8. Plopper et al, Lewin's Cells (3 rd Edition, Jones and Bartlett) <i>Elementary Biochemistry</i> 3. Berg, Tymoczko, Stryer; Biochemistry (7 th ed. 2012, Freeman) 4. Conn, Stumpf, Bruening; Principles of Biochemistry (5 th ed. 1987, Wiley)	
Learning Outcomes (not more than 100 words)	1. The students shall have a thorough understanding of the chemical components of the cell, their structure and functions. 2. Gain insight into the chemical reactions and interactions between the macromolecules regulating cellular machinery. 3. Use the knowledge gained in making scientific breakthrough in disease biology and transform global health care. 4. Useful for career in biomedical research, forensic science and pharmaceuticals.	

Course Title	ZOOMN22: Lab. exercises based on course ZOOMN21			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits- Section B		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	- The practical course aims to develop students' proficiency in basic laboratory techniques, scientific experimentation, hypothesis testing, and data interpretation while fostering good lab practices and independent problem-solving skills. - Students will master microscopy and staining techniques to analyze eukaryotic cells and their organelles. - Practical skills in cell biology will be developed through techniques like centrifugation, staining, and observing mitosis. - Students will perform biochemical tests and prepare molecular models to understand fundamental biochemical processes.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Part A: Fundamentals of Cell Biology - Drawing of the ultra structure of the cell and different organelles (from photographs provided). - Familiarization with the student's light microscope and stereobinocular microscope. - Application of centrifuge – separation of sperm from other testicular cells by low-speed centrifugation. - Diversity of eukaryotic cells – methylene blue staining of buccal epithelium, sperm, neurons, striated muscle cells; Leishman staining of mammalian blood cells. - Permeability of plasma membrane – effect of isotonic, hypotonic and hypertonic solutions on mammalian RBC. - Staining of mitochondria with Janus green in the buccal epithelium. - Mitosis in onion root tip.			15
II	Part B: Elementary Biochemistry - Preparation of models of amino acids, dipeptides and nucleotides. - Ninhydrin test for α-amino acids. - Benedict's test for reducing sugars. - Iodine test for starch. - Determination of acid value of oil.			15
Texts / References				
Learning Outcomes (not more than 100 words)	- The students must be able to perform experiments independently and become proficient in basic lab techniques. - Gain an in-depth understanding of biochemical principles, data interpretations, and logical conclusions.			

Course Title	ZOOMD21: Hormones and Diseases			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	• To define and classify hormones, neurohormones, neurotransmitters, and pheromones, and describe their characteristics. • To study the structure, biological functions, and regulation of the hypothalamus, pituitary gland, pineal gland, thyroid and parathyroid glands, adrenal glands, endocrine pancreas, and gonads. • To identify and explain hormonal dysfunctions such as diabetes insipidus, dwarfism, gigantism, acromegaly, goiter, tetany, Addison’s disease, Cushing’s syndrome, and diabetes mellitus.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Scope of Endocrinology Pituitary gland General organization and hormones Diseases: dwarfism, gigantism, acromegaly, diabetes insipidus Pineal gland General organization and hormone Sleep disorders, and jet lag			10
II	Thyroid gland General organization and hormones Diseases: goiter, myxoedema, cretinism, graves disease, Hashimoto disease Parathyroid gland General organization and hormones Diseases: bone diseases, tetany			9
III	Endocrine pancreas General organization and hormones Disease: diabetes mellitus, obesity			8
IV	Adrenal gland General organization and hormones Diseases: Addison's disease, Cushing's syndrome ,hirsutism			8
V	Testis General organization and hormones Diseases: Male infertility, Impotency, Cryptorchidism, Varicocele Ovary General organization and hormones Diseases: female infertility, polycystic ovary disease, hyperandrogenism,			10
Texts / References	1. Hadley, M.C.: Endocrinology, Prentice Hall, International Edition, 2000 2. Wilson and Foster, Williams Text Book of Endocrinology 10th edition, W.B. Saunders Company Philadelphia, 2005			
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to: 1. Understand and describe Chemical Messengers including hormones, neurohormones, neurotransmitters, and pheromones. 2. Distinguish between endocrine, paracrine, and autocrine modes of hormone delivery. 3. Analyze the Structure and Function of different Endocrine Glands: 4. Identify and analyze hormonal dysfunctions			

Course Title	ZOOSE21: Food, Preservatives and Adulterants			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		2
	Hour of Teaching (Total)	30		30
Course Objectives (not more than 100 words)	- To understand the chemistry, classification, and physico-chemical properties of key food constituents, including carbohydrates, proteins, lipids, minerals, and vitamins. - To explore the chemistry, classification, and detection methods of natural, permitted, and non-permitted preservatives. - To learn about adulteration, its types, causes, and measures, as well as modern analytical techniques for adulterant screening. - To gain knowledge of analytical methods for determining key properties and contaminants in food products, including fats, oils, dairy products, and fermented products. - To understand the detection of pesticide residues in various food items. - To study legislative regulations and safety guidelines for food products.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Chemistry, classification and physico-chemical properties of food constituents 1.1. Carbohydrate 1.2. Protein 1.3. Lipid 1.4. Minerals and Vitamins 2. Chemistry, classification and physico-chemical properties of preservatives 3. Natural, permitted and non- permitted preservatives 3.1 Methods of detection			8
II	1. Adulteration and Deterioration: Introduction, types of adulteration/substitution 2. Causes and Measure of adulteration 1.1. Sampling Procedures 1.2. Identification of substances of natural origin, pesticide residues, phototoxin and microbial contamination 1.3. Adulterant screening using modern analytical techniques			6
III	1. Determination of acid value, ester value, saponification value, iodine value, peroxide value, rancidity, moisture, ash, volatile matter, heavy metals, density, and viscosity 2. General Analytical methods for milk, milk constituents and adulterations			4
IV	1. General Analytical methods for ice cream, milk powder, butter, margarine, cheese and adulterations 2. Determination of adulteration in fats and oils			4
V	1. Analysis of fermentation products like wine, spirits, beer and vinegar 2. Determination of pesticide residues in grain, fruits, vegetables and other food items 3. Legislative regulations of food products and safety guidelines			8
Texts / References	1. Food Additives and Contaminants: Sources, Effects and Control" Editors: Alan M. Branen, P. Michael Davidson, Seppo Salminen, and John H. Thorngate Publisher: CRC Press. Edition: 2nd Edition, 2001. 2. Food Analysis "Editor: Suzanne Nielsen Publisher: Springer Edition: 5th Edition, 2017.			
Learning Outcomes (not more than 100 words)	1. Students will be able to identify and describe the chemical composition and properties of major food constituents. 2. They will understand the role and detection of various preservatives in food. 3. They will recognize different types of adulteration and apply methods to detect them. Students will acquire skills in analyzing food products for quality and safety. 4. They will comprehend food legislation and safety standards.			

Course Title		ZOOSE22: Lab. exercises based on course ZOOSE21			
Category of Course		Major / Minor / Minor (Vocational)/ SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching			Theory	Practical	Cumulative
		Credits		1	1
		Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)		• To develop practical skills in analyzing the chemical composition of food constituents. • To identify and analyze adulterants and contaminants. • To perform quality analysis of dairy products and to assess rancidity in fats and oils. • To analyze fermentation products for chemical composition. • To ensure compliance with food safety regulations.			
Course Content		The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content				Hr. of Teaching
I	Carbohydrate Analysis 1. Determination of reducing sugars using Fehling's test and Benedict's test. 2. Quantitative analysis of total sugars using the phenol-sulfuric acid method.				6
II	Protein Analysis 1. Quantitative estimation of proteins using the Kjeldahl method for nitrogen determination. 2. Protein content analysis using the Bradford or Lowry method				6
III	Lipid Analysis 1. Determination of fat content in food samples using Soxhlet extraction. 2. Measurement of acid value, saponification value, and iodine value of fats and oils.				4
IV	Quality Analysis of Dairy Products 1. Determination of fat content in milk using the Gerber method. 2. Detection of adulteration in milk using the lactometer and fat test				6
V	Rancidity Testing in Fats and Oils 1. Measurement of peroxide value to assess oxidative rancidity. 2. Detection of rancidity by determining the free fatty acid content. Food Safety and Regulatory Compliance 3. Analysis of food labels for compliance with nutritional labeling regulations. 4. Practical exercise in reading and interpreting food safety guidelines and regulations.				8
Texts / References		1. Food Additives and Contaminants: Sources, Effects and Control" Editors: Alan M. Branen, P. Michael Davidson, Seppo Salminen, and John H. Thorngate Publisher: CRC Press. Edition: 2nd Edition, 2001. 2. Food Analysis" Editor: Suzanne Nielsen Publisher: Springer Edition: 5th Edition, 2017.			
Learning Outcomes (not more than 100 words)		1. Students will be able to analyze the chemical composition of food constituents. 2. They will identify and quantify preservatives and detect adulterants and contaminants. 3. They will perform quality assessments of dairy products and determine rancidity in fats and oils. 4. Students will analyze fermentation products and understand food safety compliance.			

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	8.3.1 Chromosomal aneuploidy (Down, Turner and Klinefelter syndromes) 8.3.2 Chromosome translocation (chronic myeloid leukemia) and deletion (“cry of cat” syndrome) 8.3.3 Gene mutation (cystic fibrosis) 9. Genetic Counseling	1
Texts / References	<i>Books Recommended</i> 1. Snustad and Simmons: Principles of Genetics (2019, 7th ed. John Wiley) 2. Hartl and Jones: Essential Genetics- A Genomic Perspective (2009, Jones and Bartlet) 3. Pierce: Genetics - A Conceptual Approach (2012, Freeman) <i>Suggested Readings</i> 1. Griffith et al.: An Introduction to Genetic Analysis (Freeman 11th ed, 2015) 2. Brooker: Genetics - Analysis and Principles (2012, 4th ed McGraw Hill) 3. Russell: Genetics (2010, Benjamin Cummings) 4. Gersen & Keagle: The Principles of Clinical Cytogenetics (2005, Humana) 5. Hawley & Walker: Advanced Genetic Analysis (2003, Blackwell)	
Learning Outcomes (not more than 100 words)	• Upon successfully completing the course, the students will have a comprehensive understanding of genetic principles and mode of transmission of genetic traits. • This knowledge will prepare students for advanced studies and careers in genetics, biology, and related fields.	

Course Title	ZoomJ32: Economic Zoology			
Category of Course	Major/Minor/Minor (Vocational)/ SEC/AEC/VAC/MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- Understand the economic significance of various animals. - Explore the concepts of aquaculture, apiculture, sericulture, and animal husbandry. - Acquire knowledge in animal production and the processing of animal products.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Aquaculture 1.1 Prawn culture 1.2 Pearl Culture 1.3 Fish Culture 1.4 Fish by-products & common fish diseases			12
II	2. Sericulture 2.1 Types of silk 2.2 Silkworms and their host plants 2.3 Mulberry silkworm culture 2.4 Diseases and their control			8
III	3. Apiculture 3.1 Species of honey bees in India 3.2 Life history of <i>Apis</i> 3.3 Methods of Beekeeping 3.4 Bee products and their uses 3.5 Diseases and their control			8
IV	4. Lac culture 4.1 Lac insect and its life cycle 4.2 Cultivation of lac insect host plants, processing and uses of lac 4.3 Diseases and their control			5
V	5. Dairy and Poultry 5.1 Introduction to common dairy animals 5.2 Techniques of dairy management 5.3 Milk and milk products 5.4 Cattle diseases 5.5 Types of poultry breeds 5.6 Poultry rearing method 5.7 Poultry diseases			12
Texts / References	1. Shukla and Upadhyaya: Economic Zoology (Rastogi Publishers) 2. Omkar: Industrial Entomology (Springer Publisher) 3. Jabde: Text Book of Applied Zoology: Vermiculture, Apiculture, Sericulture, Lac culture.			
Learning Outcomes (not more than 100 words)	1. Students will gain knowledge of aquaculture systems, including breeding and post-harvest techniques. 2. Honeybee and silkworm rearing, along with the diseases affecting them, will be explored. 3. Understanding the types of poultry and dairy breeds, along with their management practices, will be emphasized.			

Course Title	ZoomJ33: Lab. exercises based on course ZoomJ31 & ZoomJ32			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	- Understand the law of segregation and genetic traits in humans. - Master various techniques using <i>Drosophila</i>. - Study the external features of economically important animals. - Perform dissections on key parts of economically significant insects. - Identify adulterations in milk and assess the quality of eggs.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	<i>Part - A: Basic Genetics</i> - Application of probability in the law of segregation with coin tossing. - Frequency of the following genetic traits in humans: widow's peak, attached ear lobe, dimple in the chin, hypertrichosis, color blindness, PTC tasting. - Study of mode of inheritance of the following traits by pedigree charts – attached ear lobe, widow's peak. - Familiarization with techniques of handling <i>Drosophila</i>, identifying males and females; Observing wild type and mutant (white eye, wingless) flies and setting up cultures. - Demonstration of the law of segregation using monohybrid crosses in <i>Drosophila</i> (using wild type and sepia-eyed flies). - Demonstration of lethal alleles using Curly (Cy) mutant in <i>Drosophila</i>. - Demonstration of multiple allelism by showing mutants of white eye series in <i>Drosophila</i>. - Study of structural chromosome aberrations (dicentric, ring chromosomes, and inversions in polytene chromosomes) from prepared slides/photographs. - Study of human karyotypes and numerical alterations (Down syndrome, Klinefelter syndrome and Turner syndrome).			30
II	<i>Part – B: Economic Zoology</i> - Dairy - External morphology and nomenclature of dairy animals. - Different types of cattle sheds and schedule of day-to-day operation of the dairy farm. - Determination of the specific gravity of milk by using a mercury lactometer. - Poultry - External morphology of poultry birds (model) - Test for fertilized and unfertilized eggs (Light test, Cracking test). - Test for egg quality (Floating test, cracking test). - Fishery - External morphology of fish. - Identification of two major carps – <i>Labeorohita</i> and <i>Catlacatla</i>. - Economically important insects - Honey Bee - Castes and life cycle (model/chart). - Mounting of the sting apparatus. - Whole mount preparation of the 3 pairs of legs. - Silkworm - Life cycle of mulberry silkworm, <i>Bombyx mori</i> - Life cycle of tasar silkworm, <i>Antheraea mylitta</i>.			30
Learning Outcomes (not more than 100)	- Students will grasp basic genetic techniques using coin tossing, public surveys, and the <i>Drosophila</i> model. - The external morphology and management of various poultry and dairy breeds, along with the quality of eggs and milk, will be studied.			

words)	3. Students will explore the life cycles of economically important insects.
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Course Title	ZOOMV31: Economic Zoology			
Category of Course	Major / Minor /Minor (Vocational)/ SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- Understand the economic significance of various animals. - Explore the concepts of aquaculture, apiculture, sericulture, and animal husbandry. - Acquire knowledge in animal production and the processing of animal products.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Aquaculture 1.1 Prawn culture 1.2 Pearl Culture 1.3 Fish Culture 1.4 Fish by-products & common fish diseases			12
II	2. Sericulture 2.1 Types of silkworms and their host plants 2.2 Mulberry silkworm culture 2.3 Diseases and their control			8
III	3. Apiculture 3.1 Species of honey bees in India 3.2 Life history of <i>Apis</i> 3.3 Methods of Beekeeping 3.4 Bee products and their uses 3.5 Diseases and their control			8
IV	4. Lac culture 4.1 Lac insect and its life cycle 4.2 Cultivation of lac insect host plants, processing and uses of lac 4.3 Diseases and their control			5
V	5 Dairy and Poultry 5.1 Introduction to common dairy animals 5.2 Techniques of dairy management 5.3 Milk and milk products 5.4 Cattle diseases 5.5 Types of poultry breeds 5.6 Poultry rearing method 5.7 Poultry diseases			12
Texts / References	1. Shukla and Upadhyaya: Economic Zoology (Rastogi Publishers) 2. Omkar: Industrial Entomology (Springer Publisher) 3. Jabde: Text Book of Applied Zoology: Vermiculture, Apiculture, Sericulture, Lac culture.			
Learning Outcomes (not more than 100 words)	1. Students will gain knowledge of aquaculture systems, including breeding and post-harvest techniques. 2. Honeybee and silkworm rearing, along with the diseases affecting them, will be explored. 3. Understanding the types of poultry and dairy breeds, along with their management practices, will be emphasized.			

Course Title	ZOOMV32: Lab. Exercises based on course ZOOMV31			
Category of Course	Major/ Minor / Minor (Vocational)/ SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• Study the external features of economically important animals. • Perform dissections on key parts of economically significant insects. • Identify milk adulterations and evaluate egg quality.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Dairy 1.1 External morphology and nomenclature of dairy animals. 1.2 Different types of cattle sheds and schedule of day-to-day operation of the dairy farm. 1.3 Determination of the specific gravity of milk by using a mercury lactometer.			8
I	2. Poultry 2.1 External morphology of poultry birds (model) 2.2 Test for fertilized and unfertilized eggs (Light test, Cracking test). 2.3 Test for egg quality (Floating test, cracking test).			6
III	3. Fishery 3.1. External morphology of fish. 3.2. Identification of two major carps – <i>Labeo rohita</i> and <i>Catla catla</i> .			6
IV	4. Economically important insects 4.1 Honey Bee i. Castes and life cycle (model/chart). ii. Mounting of the sting apparatus. iii. Whole mount preparation of the 3 pairs of legs. 4.2 Silkworm i. Life cycle of mulberry silkworm, <i>Bombyx mori</i> ii. Life cycle of tasar silkworm, <i>Antheraea mylitta</i> .			10
Learning Outcomes (not more than 100 words)	1. Students will understand the external morphology and management of various poultry, dairy breeds and quality of eggs and milk. 2. Students will understand life cycle of economically important insects.			

Course Title	ZOOMD31: Archaeozoology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	• The primary objective of this course is to offer a comprehensive understanding of the role of human and animal remains in archaeological contexts and their significance in studying past human societies. • This course, encompasses a diverse curriculum starting from an introduction to archaeozoology, exploring the Bioarchaeological chronology of South Asia, discussing significant civilizations, understanding human origins and archaic humans, to examining the evolution and utility of ancient DNA. • Alongside theoretical knowledge, a 1-credit lab exercise complements the course, emphasizing practical skills such as archaeological site visits, tool identification, dating methods, DNA extraction, and data analysis, thereby providing a holistic educational experience.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction and Archaeological Chronology of South Asia 1.1 Stone Age 1.2 Early Farming and Domestication 1.3 Bronze Age 1.4 Iron Age			7
II	Civilisations in South Asia 2.1 Sindhu-Saraswati River Valley Civilisation 2.2 Ganga-Valley civilisation			7
III	Origin of Humans 3.1 What changes have made us humans 3.2 Modern Humans and Out-of-Africa 3.3 Understanding the genetics of Peopling of South Asia			10
IV	Archaic Humans 4.1 Neanderthals 4.2 Denisovans 4.3 Genetic Introgression			5
V	Ancient DNA 5.1 Evolution of Ancient DNA Technology 5.2 Use of ancient DNA in understanding animal evolution 5.3 Genetics of Domestication and Lactose Tolerance 5.4 Archaeozoology and Diseases			10
Texts / References	1. Human Evolutionary Genetics (Second Edition): Jobling, Hollox, Hurles, Kivisild and Tyler-Smith. Garland Science, New York. 2. Neanderthal Man: In Search of Lost Genomes Svante Paabo 3. Denisovans: The Archaic Humans of the Paleolithic Period History Titans 4. Who we are and how we got here. DNA and the New Science of the Human Past. David Reich. Oxford University Press 5. Understanding Archaeology. K. Rajan 6. An Oxford Companion to Indian Archaeology. DK Chakrabarti.			
Learning Outcomes (not more than 100 words)	1. The course is pivotal for understanding the historical and biological evolution of both humans and animals within the context of South Asia's rich prehistorical timeline. 2. It explores the progression from the Stone Age to the Iron Age, delving into the major civilizations, the origin of humans, archaic humans, and the advancements in ancient DNA technology. 3. This comprehensive study sheds light on the genetic and cultural shifts that have shaped human and animal history. 4. The inclusion of tool identification, and DNA analysis, further enriches this learning experience, fostering a deeper appreciation and understanding of our ancestral past.			

Course Title	ZOOSE31: Microscopy and Histology & Histochemistry			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		3
	Hour of Teaching (Total)	30		30
Course Objectives (not more than 100 words)	To acquaint students with different types of microscopes, histological and histochemical techniques, and to get hands-on training in handling microscopes and sample preparation of various biological tissues.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Introduction to microscopy: general principles and components, numerical aperture, diffraction, and Optical Resolution			3
II	2. Light Microscopy 2.1 Bright field microscopy 2.1.1 Students microscope 2.1.2 Compound microscope 2.2 Phase contrast microscopy 2.3 Dark field microscopy			5
III	3. Sample preparation for light microscopy 3.1 Cytological technique 3.1.1 Vital Staining (Methylene blue, Gram Staining) 3.1.2 Sample fixation, staining, and mounting 3.1.3 Preparation of permanent slides 3.2 Histological and histochemical techniques 3.2.1 Fixation and fixatives 3.2.2 Tissue processing and microtomy 3.2.3 Principle and method of histological staining- Hematoxylins and eosin 3.2.4 Histochemical techniques to detect carbohydrates, proteins, and lipids			12
IV	4. Fluorescence microscopy 4.1 Epi-fluorescence microscopy 4.1.1 Source of light: halogen lamp and diodes 4.1.2 Excitation and emission spectra of various fluorophores 4.1.3 Use of filters and dichroic mirrors 4.2 Confocal microscopy 4.2.1 source of light: lasers and diodes 4.2.2 photomultiplier tubes 4.2.3 conventional filters vs digital filters 4.3 Sample preparation for Fluorescence microscopy 4.3.1 Fluorescent dyes 4.3.2. earning critical steps in immunostaining			10
Texts / References	Microscopy: 1. Karp: Cell and Molecular Biology (2007, Wiley) 2. Pollard & Earnshaw: Cell Biology (2002, Saunders) 3. Fundamentals of Light Microscopy and Electronic Imaging (2012, Douglas B. Murphy and Michael W. Davidson) 4. Ruthman: Methods in Cell Research (1970, Bell & Sons) Histology: 1. Bancroft's Theory and Practice of Histological Techniques- Suvarna et al., 2018, Elsevier health Sciences, 2018 2. Histochemistry, theoretical and applied (Vol. I, II, III) - <u>Pearse, A. G. Everson</u> (4 th ed 1980-1993,			

	Churchill- Livingstone)
Learning Outcomes (not more than 100 words)	1. Upon successful completion of the course, students will receive training in preparing various types of samples for different analyses. 2. They will become proficient in using various microscopes to analyze the samples. 3. The course will prepare students for careers in pathology, sample diagnosis, and related fields.

Course Title	ZOOSE32: Lab. Exercises based on course ZOOSE31			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	To acquaint students with different types of microscopes, histological and histochemical techniques and to get hands on training in handling microscope and sample preparation of various biological tissues.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
	Units	Course Content		Hr. of Teaching
	I	Students' Microscope and stereo-binocular basic assembly and application		4
	II	1. Tissue fixation, paraffin block preparation and sectioning. 2. Histological staining of paraffin sections using Hematoxylin and eosin 3. Histochemical staining for lipids using Sudan Black B, III and IV		12
	III	Cytological smear preparation and staining with methylene blue, Janus green and Leishman stain		6
	IV	Staining tissues with fluorescent dyes (DAPI, PI) Visualization of DNA, mitochondria and cytoskeleton under microscope using permanent slides.		8
Texts / References	<i>Microscopy:</i> 1. Karp: Cell and Molecular Biology (2007, Wiley) 2. Pollard & Earnshaw: Cell Biology (2002, Saunders) 3. Fundamentals of Light Microscopy and Electronic Imaging (2012, Douglas B. Murphy and Michael W. Davidson) 4. Ruthman: Methods in Cell Research (1970, Bell & Sons) <i>Histology:</i> 1. Bancroft's Theory and Practice of Histological Techniques- Suvarna et al., 2018, Elsevier health Sciences, 2018 2. Histochemistry, theoretical and applied (Vol. I, II, III) - <u>Pearse, A. G. Everson</u> (4 th ed 1980-1993, Churchill- Livingstone)			
Learning Outcomes (not more than 100 words)	1. Upon successful completion of the course, students will receive training in preparing various types of samples for different analyses. 2. They will become proficient in using various microscopes to analyze the samples. 3. The course will prepare students for careers in pathology, sample diagnosis, and related fields.			

Course Title	ZOOMJ41: Fundamental Endocrinology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- To define and classify hormones, neurohormones, neurotransmitters, and pheromones, and describe their characteristics. - To explain hormone delivery methods: endocrine, paracrine, and autocrine. - To understand feedback mechanisms in hormone regulation. - To describe the general mechanisms of action for peptide and steroid hormones. - To study the structure, biological functions, and regulation of the hypothalamus, pituitary gland, pineal gland, thyroid and parathyroid glands, adrenal glands, endocrine pancreas, and gonads. - To understand the roles of gastrointestinal hormones like gastrin, CCK, and secretin. - To identify and explain hormonal dysfunctions such as diabetes insipidus, dwarfism, gigantism, acromegaly, goitre, tetany, Addison's disease, Cushing's syndrome, and diabetes mellitus.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction to Endocrinology 1.1 Definition, classification and characteristics of chemical messengers (hormones, neurohormones, neurotransmitters and pheromones) 1.2 Hormone delivery: Endocrine, paracrine and autocrine modes Hormone feedback mechanisms 1.3 General mechanism of hormone action (Peptide and Steroid hormones)			10
II	Structure, biological actions and regulation of secretion of 2.1 Hypothalamus and Pituitary gland 2.2 Pineal gland 2.3 Thyroid and Parathyroid gland 2.4 Adrenal gland 2.5 Endocrine pancreas			20
III	Reproductive hormones: 3.1 Structural organization of gonads 3.2 Synthesis, biological action and regulation of secretion of 3.2.1 Estrogen 3.2.2 Progesterone 3.2.3 Testosterone			6
IV	Gastrointestinal hormones: Gastrin, CCK and Secretin			3
V	Hormonal dysfunctions: 4.1 Diabetes insipidus 4.2 Dwarfism, gigantism, acromegaly 4.3 Goitre 4.4 Tetany 4.5 Addison's disease 4.6 Cushing's syndrome 4.7 Diabetes mellitus (Type I and II)			6
Texts / References	1. Hadley: Endocrinology, Fifth Edition, 2000, Prentice Hall 2. Norris: Vertebrate Endocrinology, Fourth Edition, 2007, Academic Press 3. Goodman: Basic Medical Endocrinology, Fourth Edition, 2009, Elsevier Inc.			
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to: 1. Understand and Describe Chemical Messengers including hormones, neurohormones, neurotransmitters, and pheromones. 2. Distinguish between endocrine, paracrine, and autocrine modes of hormone delivery. 3. Describe hormone feedback mechanisms and their role in maintaining physiological balance. 4. Understand the general mechanisms of action for peptide and steroid hormones, including receptor interactions and signal transduction pathways. 5. Analyze the Structure and Function of different Endocrine Glands: 6. Identify and Analyze Hormonal Dysfunctions:			

Course Title	ZOOMJ42: Developmental Biology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above):			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- To provide historical background about the key hypothesis and discoveries in the field of embryology - To know about the processes by which a single cell (zygote) formed after the fertilization becomes an embryo and then an adult organism. - To provide knowledge about the crucial developmental processes, including gametogenesis, fertilization, cleavage, and gastrulation - To explain the concept of commitment (specification and determination), competence, differentiation, regeneration. - To understand the role of primary organizer and axis formation in frog by experiments performed by Hans Spemann, Pieter Nieuwkoop, and Gimlich and Gerhart			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Historical perspective and scope of developmental biology 1.1 Contributions of several scientists 1.2 Anatomical, Experimental and Genetic Approaches 1.3 Karl Ernst von Baer Principles			3
	2. Gametogenesis 2.1 Spermatogenesis 2.2 Oogenesis			5
II	3. Fertilization 3.1 External fertilization: sea urchin 3.2 Internal fertilization			5
	4. Cleavage 4.1 Patterns and types 4.2 Cleavage in frog and chick embryo			3
III	5. Fate map and gastrulation in frog and chick 5.1 Fate map 5.2 Gastrulation in frog 5.3 Gastrulation in chick			5
	6. Concept of commitment (specification and determination), competence and differentiation 6.1. Autonomous, Conditional, and Syncytial specification 6.2 Induction and Competence 6.3 Definition and types of Differentiation			7
IV	7. Primary organizer and axis formation in frog 7.1 Experiments of Hans Spemann 7.2 Nieuwkoop center 7.3 Experiments of Pieter Nieuwkoop, and Gimlich and Gerhart			7
	8. Extra embryonic membranes			2
V	9. Hormonal control of metamorphosis in frog 10. Concept of regeneration 10.1 Morphallaxis 10.2 Epimorphosis 10.3 Stem cell-mediated regeneration 10.4 Compensatory regeneration			4
Texts / References	1. Gilbert: Developmental Biology (8 th ed., 2006, Sinauer) 2. Balinsky: An Introduction to Embryology (1981, CBS) 3. Wolpert: Principles of Development (3 rd ed., 2007, Oxford) 4. Sastry and Shukla: Developmental Biology (2 nd ed., 2021, Rastogi Publications) 5. Verma and Agarwal: Chordate Embryology (2010, S. Chand)			

Learning Outcomes (not more than 100 words)	After studying the course, the student will: 1. Comprehensively understand developmental biology from embryo to adult 2. Understand the fundamental concepts of commitment (specification and determination), competence, differentiation, metamorphosis, and regeneration 3. Understand the importance of primary organizer in the embryonic development			
Course Title	ZOOMJ43: Evolution			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above):			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		2
	Hour of Teaching (Total)	30		30
Course Objectives (not more than 100 words)	- Studying evolutionary biology aims to understand the processes driving the diversity and adaptation of life on our planet. - Key objectives include exploring the mechanisms of evolution, such as natural selection, genetic drift, and gene flow, and reconstructing the evolutionary relationships among species through phylogenetic analysis. - Students learn to interpret fossil records and molecular data to trace evolutionary history and identify patterns of speciation and extinction. - The field of evolution also examines the role of evolutionary processes in shaping ecological interactions and behaviors.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. History of Evolutionary Thoughts 2. Evidence of organic evolution 2.1 Comparative anatomy 2.2 Comparative embryology 2.3 Palaeontology 2.4 Biochemistry and genetics 2.5 Zoogeography 3. Theories of Evolution 3.1 Lamarckism 3.2 Darwinism 3.3 Concept of synthetic theory			01 03 03
II	4. Population Genetics 4.1 Gene frequencies in Mendelian populations 4.2 Hardy- Weinberg equilibrium 4.3 Evolutionary forces (Mutation, Selection, Random genetic drift, Migration)			06
III	5. Genetic polymorphism 5.1 Chromosomal 5.2 Allozyme 5.3 DNA polymorphism			03
IV	6. Modes of speciation 7. Isolating mechanisms 8. Evolution at molecular level			02 02 02
V	9. Evolution through ages (Geological columns; Evolution of horse; Evolution of man)			04
Texts / References	1. Futuyma: Evolutionary Biology (2005, Sinauer) 2. Hall and Hallgrímsson: Strickberger’s Evolution (2008, Jones and Bartlet) 3. Hartl and Clark: Principles of Population Genetics (1989 & 1997, Sinauer) 4. Moody: Introduction to Evolution (1978, Kalyani). 5. Rastogi: Organic Evolution (2007, Kedarnath & Ramnath).			
Learning Outcomes (not more	- The aspect of evolutionary biology helps to understand the mechanism of evolution of different forms of life on our earth planet. - How genetic and ecological factors facilitate the process of speciation and the way organisms adapt to their environments over time could be known by going through the study			

than 100 words)	of evolutionary biology. 3. One can learn to construct and analyze evolutionary trees to understand the relationships between species. 4. Its knowledge helps to examine changes in DNA sequences to infer evolutionary relationships among different forms of life.
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Course Title	ZOOMJ44: Systematics and Taxonomy			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above):			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		2
	Hour of Teaching (Total)	30		30
Course Objectives (not more than 100 words)	• The study of Systematics and Taxonomy aims to understand and organize biological diversity by classifying organisms into hierarchical categories based on evolutionary relationships. • This knowledge facilitates the construction of phylogenetic trees, aids in identifying and naming species, and supports accurate communication within the scientific community. • It also enhances research by providing a framework for data organization, informs conservation efforts by identifying endangered species, and integrates insights from various biological disciplines.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction to taxonomy and its relationship with systematics Zoological nomenclature 1. Binominal 2. Trinominal			05
II	Kinds of taxonomic characters 1. Morphological 2. Embryological 3. Behavioural 4. Ecological 5. Numerical			08
III	Kinds of zoological classification 1. Components of classification 2. Linnaean hierarchy			04
IV	Concept of species 1. Typological 2. Nominalistic 3. Biological 4. Evolutionary			06
V	Molecular Taxonomy			03
Texts / References	<i>Books Recommended</i> 1. Boolotian and Stiles: College Zoology (1981, Macmillan) 2. Dalela and Sharma: Animal Taxonomy and Museology (1976, Jai Prakash Nath). 3. Kapoor: Theory and Practicals of Animal Taxonomy (1988, Oxford & IBH). 4. Mayr and Ashlock: Principles of Systematic Zoology (1991, McGraw Hill). 5. Roymahoney: Laboratory Techniques in Zoology (1966, Butterworths). 6. Simpson: Principles of Animal Taxonomy (1962, Oxford).			
Learning Outcomes (not more than 100 words)	Upon completing this course/module, students will: 1. Understand the core principles of taxonomy and systematics, including their interrelationship in biological classification. 2. Accurately apply zoological nomenclature rules, distinguishing between binomial and trinomial naming systems. 3. Identify and utilize various taxonomic characters in classifying organisms. 4. Evaluate different species concepts (typological, nominalistic, biological, and evolutionary) and their relevance in modern taxonomy, along with the role of molecular techniques in elucidating phylogenetic relationships.			

Course Title	ZoomJ45: Lab. exercises based on course ZoomJ41 & ZoomJ42			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	<i>Fundamental Endocrinology:</i> - This course equips students with proficiency in handling and maintaining laboratory rodents, surveying endocrine glands and sex organs, and conducting vaginal smear analysis to study the estrous cycle. - It covers surgical techniques like ovariectomy and orchidectomy, histological analysis of endocrine and reproductive organs, and sperm parameter assessment. - Additionally, students will learn hormone estimation using the ELISA technique. <i>Developmental Biology:</i> - Students will learn to identify animal eggs based on yolk distribution, understand meiotic oogenesis, recognize sperm structure, and study embryonic developmental stages in frogs and chicks through models and slides.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1.1 Handling, sexing, numbering and maintenance of rat/mouse. 1.2 Study of the estrous cycle by vaginal smear preparation.			10
II	2.1 General survey of endocrine glands and accessory sex organs in rat/mouse. 2.2 Demonstration of Ovariectomy and Orchidectomy in rat 2.3 Study of sperm parameters (Sperm count, motility and viability). 2.4 Estimation of any hormone by ELISA (Demonstration only)			10
III	Study of histological slides of the following: 1. Pituitary 2. Thyroid and parathyroid 3. Endocrine, pancreas 4. Adrenal 5. Testis 6. Ovary 7. Uterus 8. Seminal vesicle			10
IV	1. Study of different types of eggs (snail, <i>Drosophila</i> , frog, chick and mouse) 2. Demonstration of sperm motility in rat/mice 3. Study of meiotic oogenesis by dissecting and observing ovaries of <i>Drosophila</i>			10
V	4. Study of eggs and tadpoles of frogs from collected/preserved material 5. Study of frog development through models 6. Window preparation on hen's egg 7. Demonstration of chick embryonic development, making window preparation of fertilized egg 8. Study of whole mount preparations of chick embryos at 16-18, 24-28, 33-36 and 42-48 hours of development			20
Texts / References	1. Advanced Practical Zoology by PS Verma & PC Srivastava . S. Chand Publishing, 2015			
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to: - Demonstrate proper techniques for handling, sexing, numbering, and maintaining rats/mice. - Identify and describe the major endocrine glands and accessory sex organs in rats/mice. - Analyze the estrous cycle in rodents through vaginal smear preparation. - Understand and observe the procedures for ovariectomy and orchidectomy in rats. - Identify and interpret histological slides of various endocrine glands and reproductive organs (pituitary, thyroid, parathyroid, endocrine pancreas, adrenal, testis, ovary, uterus, seminal vesicle). - Assess sperm parameters, including count, motility, and viability. - Understand the principles of hormone estimation using ELISA and observe its demonstration.			

Course Title	ZoomJ46: Lab. exercises based on course ZoomJ43 & ZoomJ44			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	<i>Evolution:</i> - This course explores the processes driving life's diversity and adaptation, focusing on mechanisms like natural selection, genetic drift, and gene flow. - Students learn to reconstruct evolutionary relationships through phylogenetic analysis, interpret fossil records and molecular data, and trace patterns of speciation and extinction. - Additionally, the course examines how evolutionary processes shape ecological interactions and behaviors. <i>Systematics and Taxonomy:</i> - Students will learn to classify organisms based on evolutionary relationships, aiding in constructing phylogenetic trees, identifying and naming species, and facilitating scientific communication. - This knowledge also supports research and informs conservation efforts.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	<i>Part – A: Evolution (Credit 1)</i> 1. Natural selection – experiment using red and white-eyed Drosophila flies. 2. Isolating mechanism - Demonstration of sexual isolation in Drosophila: <i>D. bipectinata</i> and <i>D. malerkotliana</i> . 3. Calculation of Hardy-Weinberg equilibrium in human population considering MN and ABO blood group system. 4. Fossils – One representative fossil each from Foraminifera, Trilobita, Ammonites and Echinodermata. 5. Models of chordate fossils – Brontosaurus, Dimetrodon, Archaeopteryx, Dinoceras. 6. Evolution of horse – through models.			30
II	<i>Part – B: Systematics and Taxonomy (Credit 1)</i> 1. Identification of five locally available fishes based on their morphological characters. 2. Identification of the following species of Drosophila based on the pattern of their sex combs: <i>D. melanogaster</i> , <i>D. ananassae</i> , <i>D. bipectinata</i> and <i>D. malerkotliana</i> . 3. Zoological names of some common animals. 4. Preservation of insects. 5. Fixation of freshwater protozoans.			30
Texts / References				
Learning Outcomes (not more than 100 words)	1. Students will conduct natural selection experiments with Drosophila and calculate Hardy-Weinberg equilibrium, demonstrating their understanding of evolutionary processes. 2. The ability to identify and classify organisms, including locally available fishes and different Drosophila species, will be developed through hands-on practice. 3. Practical experience in preserving insects, fixing freshwater protozoans, and recognizing key fossils will enhance their understanding of evolutionary history and taxonomy.			

Course Title	ZOOMN41: Fundamental Endocrinology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- To define and classify hormones, neurohormones, neurotransmitters, and pheromones, and describe their characteristics. - To explain hormone delivery methods: endocrine, paracrine, and autocrine. - To understand feedback mechanisms in hormone regulation. - To describe the general mechanisms of action for peptide and steroid hormones. - To study the structure, biological functions, and regulation of the hypothalamus, pituitary gland, pineal gland, thyroid and parathyroid glands, adrenal glands, endocrine pancreas, and gonads. - To understand the roles of gastrointestinal hormones like gastrin, CCK, and secretin. - To identify and explain hormonal dysfunctions such as diabetes insipidus, dwarfism, gigantism, acromegaly, goitre, tetany, Addison's disease, Cushing's syndrome, and diabetes mellitus.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction to Endocrinology 1.2 Definition, classification and characteristics of chemical messengers (hormones, neurohormones, neurotransmitters and pheromones) 1.2 Hormone delivery: Endocrine, paracrine and autocrine modes Hormone feedback mechanisms 1.4 General mechanism of hormone action (Peptide and Steroid hormones)			10
II	Structure, biological actions and regulation of secretion of 2.1 Hypothalamus and Pituitary gland 2.2 Pineal gland 2.3 Thyroid and Parathyroid gland 2.4 Adrenal gland 2.5 Endocrine pancreas			20
III	Reproductive hormones: 3.1 Structural organization of gonads 3.2 Synthesis, biological action and regulation of secretion of 3.2.1 Estrogen 3.2.2 Progesterone 3.2.3 Testosterone			6
IV	Gastrointestinal hormones: Gastrin, CCK and Secretin			3
V	Hormonal dysfunctions: 4.1 Diabetes insipidus 4.2 Dwarfism, gigantism, acromegaly 4.3 Goitre 4.4 Tetany 4.5 Addison's disease 4.6 Cushing's syndrome 4.7 Diabetes mellitus (Type I and II)			6
Texts / References	1. Hadley: Endocrinology, Fifth Edition, 2000, Prentice Hall 2. Norris: Vertebrate Endocrinology, Fourth Edition, 2007, Academic Press 3. Goodman: Basic Medical Endocrinology, Fourth Edition, 2009, Elsevier Inc.			
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to: 1. Understand and Describe Chemical Messengers including hormones, neurohormones, neurotransmitters, and pheromones. 2. Distinguish between endocrine, paracrine, and autocrine modes of hormone delivery. 3. Describe hormone feedback mechanisms and their role in maintaining physiological balance. 4. Understand the general mechanisms of action for peptide and steroid hormones, including receptor interactions and signal transduction pathways. 5. Analyze the Structure and Function of different Endocrine Glands: 6. Identify and Analyze Hormonal Dysfunctions:			

Course Title	ZOOMN42: Lab. exercises based on course ZOOMN41			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• The course develops skills in handling laboratory rodents, surveying endocrine glands and sex organs, and analyzing vaginal smears to study the estrous cycle. • To understand the surgical techniques and implications of ovariectomy and orchidectomy in rats. • To gain skills in identifying and analyzing histological slides of endocrine glands and reproductive organs, including the pituitary, thyroid, parathyroid, pancreas, adrenal, testis, ovary, uterus, and seminal vesicle. • To measure and assess sperm parameters such as count, motility, and viability. • To observe and comprehend the principles of hormone estimation using the ELISA technique.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1.1 Handling, sexing, numbering and maintenance of rat/mouse. 1.2 Study of the estrous cycle by vaginal smear preparation.			10
II	2.1 General survey of endocrine glands and accessory sex organs in rat/mouse. 2.2 Demonstration of Ovariectomy and Orchidectomy in rat 2.3 Study of sperm parameters (Sperm count, motility and viability). 2.4 Estimation of any hormone by ELISA (Demonstration only)			10
III	3.1 Study of histological slides of the following: 3.2 Pituitary 3.3 Thyroid and parathyroid 3.4 Endocrine, pancreas, 3.5 Adrenal 3.6 Testis & Ovary 3.7 Uterus 3.8 Seminal vesicle			10
Texts / References	1. Advanced Practical Zoology by PS Verma & PC Srivastava . S. Chand Publishing, 2015			
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to: 1. Demonstrate proper techniques for handling, sexing, numbering, and maintaining rats/mice. 2. Identify and describe the major endocrine glands and accessory sex organs in rats/mice. 3. Analyze the estrous cycle in rodents through vaginal smear preparation. 4. Understand and observe the procedures for ovariectomy and orchidectomy in rats. 5. Identify and interpret histological slides of various endocrine glands and reproductive organs (pituitary, thyroid, parathyroid, endocrine pancreas, adrenal, testis, ovary, uterus, seminal vesicle). 6. Assess sperm parameters, including count, motility, and viability. 7. Understand the principles of hormone estimation using ELISA and observe its demonstration.			

Course Title	ZOOMJ51: Functional Anatomy of Non-chordates			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- The course would help students to understand the characteristic morphological and anatomical features of diverse animal groups. - Will help students to develop basic understanding of structure-function relationships and its adaptive significance in survival in different habitats. - Students will be able to outline the ecological, medicinal and economic significance of various life forms.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Protozoa Study of <i>Euglena</i> and <i>Monocystis</i> (locomotion, nutrition and reproduction)			4
	2. Organization of Metazoans 2.1 Grades of organization 2.2 Symmetry, Germ layers, diploblastic and triploblastic organization, coelom, metamerism			5
	3. Porifera 3.1 Study of <i>Leucosolenia</i> and <i>Sycon</i> (structure, skeleton and reproduction) 3.2 Aquaferous system			3
II	Cnidaria and Ctenophora 4.1 Study of <i>Obelia</i> and <i>Aurelia</i> (structure and reproduction) 4.2 Salient features of ctenophores and comparison with cnidarians			7
III	5. Platyhelminthes: <i>Fasciola</i> and <i>Taenia</i> (structure, reproduction, life-cycle and parasitic adaptations)			5
	6. Nematelminthes: <i>Ascaris</i> - (Structure, reproduction and life-cycle)			3
IV	7. Annelida: <i>Nereis</i> - (Structural features and reproduction)			3
	8. Arthropoda 8.1 <i>Palaemon</i> (Structural features and reproduction), 8.2 Types of metamorphosis in insects 8.3 Salient features and evolutionary significance of Onychophora			4
V	9. Mollusca 9.1 <i>Unio</i> : Structural features and reproduction 9.2 Torsion and detorsion in gastropods			6
	10. Echinodermata 10.1 <i>Asterias</i> : Structural features and reproduction			5
	10.2 Hydrostatic system			
Texts / References	1. Barnes: Invertebrate Zoology (Recent edition, Holt Saunders International) 2. Marshall: Parker & Hashwell Textbook of Zoology, Vol I (Recent edition, Macmillan) 3. Barnes: The Invertebrates – A Synthesis (Recent edition, Blackwell) 4. Hunter: Life of Invertebrates (Recent edition, Collier Macmillan) 5. Moore: An Introduction to the Invertebrates (2001, Cambridge University Press)			
Learning Outcomes (not more than 100 words)	1. The course will empower students to appreciate the diversity in life forms and their ecological, economic and medicinal importance in human life. 2. Will encourage students to critically analyze the organizational complexity and characteristic features of non-chordates making them familiarize with the morphology and anatomy of representatives of various animal phyla.			

Course Title	ZOOMJ52: Animal Behaviour			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		2
	Hour of Teaching (Total)	30		30
Course Objectives (not more than 100 words)	- To introduce students to the patterns of behaviour and core animal behaviour theories. - To identify some of the mechanisms involved in the production of a behavior by an animal. - To study and understand the ecological and evolutionary perspectives for the diversity of behaviour.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Introduction and Patterns of behaviour			4
II	2. Instinct and learning 2.1 Innate behavior 2.2 Learned behavior and types of learning			4
III	3. Control of behavior 3.1 Neural control 3.2 Hormonal control 3.3 Genetic and environmental control			8
IV	4. Social organization 4.1 Communication: Channels and Functions 4.2 Foraging and antipredator behavior 4.3 Mating system and Parental care 4.4 Group living, altruism and cooperation 4.5 Dispersal and migration 4.6 Coordination and orientation			10
V	5. Biological rhythms			4
Texts / References	1. Lee Alan Dugatkin (2020): Principles of Animal Behavior (4 th ed., University of Chicago Press). 2. Drickamer, Vessey & Jakob (2002): Animal Behaviour – Mechanisms, Ecology and Evolution (5 th ed., McGraw-Hill). 3. Judith Goodenough, Robert A. Wallace, Betty McGuire (2010): Perspectives on Animal Behavior (3 rd ed., John Wiley & Sons). 4. David McFarland (2009) Animal Behaviour: Psychobiology, Ethology and Evolution (3 rd ed., Longman Scientific).			
Learning Outcomes (not more than 100 words)	1. The student will be able to understand the fundamental concepts of ethology that apply to most of the species. 2. The student will be able to predict the proximate (physiology, cognition, etc) and ultimate (evolution, adaptation) explanations of behaviour. 3. Students will have better analyses and interpretation skills while observing and animal. 4. Students will be able to relate animal behavior with other aspects like animal biodiversity, conservation and ecology.			

Course Title	ZOOMJ53: Immunology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above):			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- To provide historical perspective along with basic understanding about immunity. - To provide knowledge about immune cells and lymphoid organs - To provide knowledge on properties of antigens, and structure, types, and function of antibodies - To acquire knowledge about complement system and their role in immunity. - To explain the concept of hypersensitivity, auto immunity and immune deficiencies			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content	Hr. of Teaching		
I	1. Introduction to Immunology 1.1 Historical perspective 1.2 Clonal Selection Theory 1.3 Types of Immunity 2. Cells and organs of the immune system 2.1. Hematopoiesis 2.2. Hematopoietic stem cells and progenitor cells 2.3. Types of immune cells: lymphoid and myeloid 2.4. Primary and secondary lymphoid organs and lymphatic system	4 8		
II	3. Innate Immunity 4. Humoral Immunity 4.1. Antigen and immunogen 4.2. Hapten 4.3. Adjuvant 4.4. Immunoglobulins: types, structure, function, and diversity 4.5. Structure of B cell receptor 4.6. Antibody-mediated effector functions 4.7. Antigen-Ab reactions for immunological techniques	4 8		
III	5. Complement System 5.1 Complement components 5.2 Classical pathway 5.3 Alternative pathway 5.4 Lectin Pathway 6. Cell-mediated Immunity 6.1. Structure of T cell receptor 6.2. Structure and organization of Major Histocompatibility Complex and T cell interaction 6.3. Antigen processing and presentation	3 9		
IV	7. General Introduction to Immunological Disorders 7.1 Autoimmunity 7.2 Immunodeficiency 7.3 Hypersensitivity 7.4 Tolerance 8. Concept of vaccination	5 4		
Texts / References	1. Janeway's Immunobiology, 7 th ed. 2008, Garland Science Publication. 2. Kubeyet <i>al.</i> : Immunology, 7 th ed. 2013, W.H. Freeman and Company Publication, New York. 3. Roitt and Delvis: Roitt's Essential Immunology, 6 th ed. 2006, Blackwell Publication 4. Fundamental Immunology, 6 th ed. 2008, Lippincott Williams & Wilkins, Philadelphia 5. Acharya <i>et al.</i> : Immunology, 2 nd ed. 2011, Kalyani Publishers, Ludhiyana, Punjab			
Learning Outcomes (not more than	1. Understand about the immune system that how an intricate system regulates the immune response against a vast variety of pathogen. 2. Understand how immune armaments get formed and mature in our body, what are the organs			

100 words)	that are participating in their formation and maturation. 3. Understand the basic structure and function of immunoglobulin generation and their classification as well as their function in immune responses 4. Understand the role of complement system 5. Able to know how MHC molecules and T cells work together and protect from various diseases 6. Know about the different approaches of conception of vaccination. Different method of the vaccine preparation, its characteristic properties, uses, schedule of vaccination of different age groups			
Course Title	ZOOMJ54: Biology of Infectious Disease			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above):			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		2
	Hour of Teaching (Total)	30		30
Course Objectives (not more than 100 words)	• Provide information on infectious agents and types of pathogens. • Educate on parasites, parasitism, vectors, and zoonotic diseases. • Explain pathogenesis and diseases caused by protozoa and helminths.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Introduction to infectious agents 1.1 Animal associations 1.2 Concept of harms 1.3 Signs and symptoms 2. Types of pathogens and diseases 2.1 Viruses- Influenza, Dengue 2.2 Bacteria - Mycobacteria, Salmonella 2.3 Fungi - Candida, Histoplasma 2.4 Parasites- Protozoans and Helminths			2 8
II	3. Pathogenicity and Infection 1.1 Pathogenicity and Infectious Disease 1.2 Virulence 1.3 Exposure and Transmission 4. General concept of parasites and parasitism 4.1 Introduction to parasitology: basic principles and concepts 4.2 Systematics, ecology and evolution 4.3 Vector 4.4 Zoonosis			3 4
III	5. Parasitic protozoans: life cycle, pathogenesis and diseases caused by - 5.1 Entamoeba histolytica 5.2 Plasmodium 5.3 Toxoplasma gondii			4
IV	6. Helminths: life cycle, pathogenesis and diseases caused by - 6.1 <i>Paragonimus westermani</i> 6.2 <i>Trichinella spiralis</i> 7. Dynamics of Host-Parasite interaction: Evasive strategies			3 2
Texts / References	1. Smith, Animal Parasitology 1996. Cambridge University Press. 2. Chandler and Reid, Introduction to Parasitology 1970, Wiley. 3. Marr et al., Molecular Medical Parasitology 2003, Elsevier. 4. <u>Ash and Orihel, Atlas of Human Parasitology</u> . American Society for Clinical Pathology Press 5 th edition, 2007. 5. Willey et al. Prescott's Microbiology, McGraw-Hill Higher Education, 9 th edition, 2011			
Learning Outcomes	• Students will learn about infectious agents, types of pathogens, pathogenicity, and infection. • Also, they will come to know about parasites and parasitism, vector, and zoonotic disease.			

(not more than 100 words)	They will be familiar with pathogenesis and diseases caused by protozoan and helminths.		
Course Title	ZOOMJ55: Lab. exercises based on course ZOOMJ51 & ZOOMJ52		
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)		
Credits & Hour of Teaching		Theory	Practical
	Credits		2
	Hour of Teaching (Total)		60
Course Objectives (not more than 100 words)	- To impart concepts and knowledge about the morphology, and anatomy of non-chordates through observations of real animals/preserved specimens/models. - To observe the diversity in animals making students understand about their distinguishing features. - To enhance the student's understanding of animal behavior through observations, analyses, and interpretation of several types of animal behavior. - To highlight the adaptive significance of the behavior to the organisms.		
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.		
Units	Course Content	Hr. of Teaching	
I	<i>Part A: Functional Anatomy of Non-chordates (Credit 1)</i> I. Protozoa to Pseudocoelomates - Protozoa: Culturing of <i>Paramecium</i> and <i>Euglena</i> and observation of their locomotion - Porifera - Study of canal systems (asconoid, syconoid, leuconoid) from prepared slides and models - Study of the following museum specimens: <i>Euplectella</i>, <i>Hyalonema</i>, <i>Cliona</i>, <i>Euspongia</i> - Cnidaria - Study of the following through prepared slides: T.S. of Hydra through ovary and testis, Scyphistoma and Ephyra. - Study of the following museum specimens: <i>Physalia</i>, <i>Porpita</i>, <i>Tubipora</i>, <i>Corallium</i>, <i>Gorgonia</i>, <i>Fungia</i>, <i>Millepora</i>. - Ctenophora: Study of the following museum specimens: <i>Hormiphora</i>, <i>Beroe</i> - Platyhelminthes: Study of the following from slides/specimen: <i>Dugesia</i>, <i>Polystomum</i>, <i>Schistosoma</i>, <i>Echinococcus</i>, <i>Fasciola</i>, <i>Taenia</i>, <i>Miracidium</i>, sporocyst, redia, cercaria, metacercaria, Hexacanth bladder worm - Nemathelminthes: Study of the following from slide/specimen: <i>Ascaris</i> (male and female)	15	
II	II. Coelomates - Annelida - Leech - External features - Dissections: Digestive system - Mounting of jaws and salivary glands <i>Nereis</i>: Permanent stained preparation of parapodium - Study of the following museum slides/specimens: <i>Nereis</i>, <i>Aphrodite</i>, <i>Arenicola</i>, <i>Sabella</i>, <i>Acanthobdella</i>, <i>Branchellion</i>, <i>Bonellia</i> - Trochophore Larva, T.S. of Leech through crop sac - Arthropoda - Study of the following museum specimens/slides: <i>Daphnia</i>, <i>Cypris</i>, <i>Cyclops</i> (male and female), <i>Lepas</i>, <i>Balanus</i>, <i>Cancer</i>, <i>Julus</i>, <i>Scolopendra</i>, <i>Lepisma</i>, Cricket, grasshopper, termite, <i>Apis</i>, wasp, butterfly, moth. - Study of the following larval forms from slides: Nauplius, Zoea and Megalopa. - Mollusca: Study of the following museum specimens: <i>Chiton</i>, <i>Cyprea</i>,	15	

	<i>Patella, Aplysia, Doris, Dentalium, Mytilus, Pecten</i> <i>Teredo, Sepia, Loligo, Octopus.</i> 10. Echinodermata 10.1 <i>Asterias, Echinus, Holothuria, Cucumaria, Clypeaster</i> (cake urchin) 10.2 Study of the following larval forms from slides: Bipinnaria, Brachiolaria, Auricularia, Ophiopluteus and Echinopluteus.	
III	<i>Part B: Animal Behaviour (Credit 1)</i> 1. Study of habitat selection in the first and third instar larvae of <i>Drosophila melanogaster</i>. 2. Habituation in earthworms/mosquito larvae 3. Locomotor behaviour of dipteran larvae (Housefly/blowfly/fruit fly): 3.1. Locomotion on different types of substrata (writing paper, plastic sheet and sandpaper) 3.2. Effects of light intensity and light quality on the rate of locomotion 4. Study of individual and social behavioural patterns of a troop of monkey 5. Study of interspecific association between cattle and egrets / mixed flock foraging in birds. 6. Study of circadian rhythm by measuring locomotor activity patterns in mice (demonstration). 7. Study of memory and learning in mice through maze-based experiments.	30
Texts / References	1. SS Lal: Practical Zoology- Invertebrates (12th ed., Rastogi Publications) 2. PS Verma: A manual of Practical Zoology-Invertebrates (11th ed., S Chand publications)	
Learning Outcomes (not more than 100 words)	1. Students will get familiarized with the morphology and anatomy of representatives of various animal phyla. 2. Museum specimen studies will provide students a comprehensive idea of structural features of non-chordates and the basis of classification. 3. Students will have enhanced Understanding of the types of animal behavior and their importance to the organisms. 4. Students will be able to relate animal behavior with other subjects such as biodiversity, evolutionary biology, ecology, conservation biology and genetic basis of the behavior.	

Course Title	ZOOMJ56: Lab. exercises based on course ZOOMJ53 & ZOOMJ54			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	(a) To provide a basic understanding about the use of mice for immunological experiments along with precautions being considered during their handling (b) To learn cell counting by hemocytometer (c) To identify immune cells on the basis of nuclear morphology (d) To learn the staining procedure of the protozoans and helminth parasites			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. An introduction and acquaintance with necessary precautions in immunological practical and proper handling of the mice. 2. Study of different lymphoid organs, counting and fix the cell number for particular experiments.			10
II	1. To perform retro-orbital bleeding with the help of capillary tubes and preparation of blood film for differential cell count. 2. Separation of peritoneal macrophages and their identification on the basis of non specific esterase staining.			10
III	1. Enumeration of apoptotic cells by trypan blue exclusion test. 2. To study antigen-antibody interaction with the help of Mancini/Ouchterlony method Immunodiffusion).			10
IV	1. Introduction on collection and preservation of specimens, quality control and safety measures. 2. Staining procedure for protozoans and helminth parasites. 3. Study of bacteria by Gram stain. 4. Preparation of thin blood smear: total and differential counting. 5. Identification of protozoan parasites from permanent slides. <i>Trypanosoma</i> : epimastigote or trypomastigote form. <i>Leishmania</i> : promastigote and amastigote form. <i>Plasmodium</i> : sporozoites and signet ring. <i>Toxoplasma</i> : tachyzoites and bradyzoites. <i>Giardia</i> <i>Entamoeba</i> : trophozoites 6. Identification and characterization of helminth parasites from permanent slides <i>Cercaria</i> of <i>Fasciola</i> . Eggs of <i>Schistosoma</i> . Cyst of <i>Echinococcus granulosus</i> . Microfilariae of <i>Wuchereria</i> . 7. Parasite culture (demonstration only)			30
Learning Outcomes (not more than 100 words)	After studying the course, the student will: (a) Learn about mice handling (b) Able to identify and locate primary and secondary lymphoid organs (c) Learn the methods for the isolation and characterization of immune cells, including macrophages (d) Able to perform TC/DC in blood			

Course Title	ZOOIN57: Internship			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD / Internship / Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	2
	Hour of Teaching (Total)		2 weeks	2 weeks
Course Objectives (not more than 100 words)	(a) To provide opportunity for career exploration and to learn new skills in industry/organization/research labs with faculty and researchers in their own or other curriculum. (b) To provide opportunity to utilize their learned skills in the class room into practice and to enhance their skills in the research environment/field work and benefit from the Mentor/Supervisor and developing a learning network.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Visit to a research lab/industry related to the curriculum and submission of a report.			2 weeks
Learning Outcomes (not more than 100 words)	(a) Learn to apply theoretical knowledge into research work/field work in a research lab or industry. (b) Learn to prepare a scientific report. (c) Learn to develop/use a method to study an objective and enhance their skill to interact with fellow members/Mentor.			

Course Title	ZOOMV51: Aquaculture			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	• Demonstrate the types and practices of aquaculture. • Understand the physicochemical requirements of aquaculture systems. • Learn the nutritional needs of species and their disease management. • Gain knowledge of advanced omics approaches in aquaculture. • Comprehend the socio-economic status of fish farmers.			
Course Content	The course content is divided into minimum 3 Units to maximum 5 units along with defined hours of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Overview of aquaculture 1.1 Scope and definition 1.2 Introduction to finfish and shellfish 1.3 Aquaculture systems and methods 2. Water quality and aquatic health 2.1 Physicochemical conditions of pond water 2.2 Causes and effects of water pollution			10
II	3. Freshwater Aquaculture 3.1 Fish culture 3.1.1 Carp culture 3.1.2 Ornamental fish culture 3.2 Prawn culture 3.3 Pearl culture 3.4 Integrated fish farming			10
III	4. Fish Feed formulation and nutrition 4.1 Supplementary 4.2 Medicinal 4.3 Artificial 5. Fish disease and management 5.1 Microbial 5.2 Parasitic 5.3 Nutritional deficiency			7
IV	6. Fish by-products 6.1 Pharmaceuticals 6.2 Enzymatic and biomolecular materials 6.3 Commercial products 7. Preservation methods in aquaculture			6
V	8. Aquaculture biotechnology 8.1 Genetic applications in aquaculture 8.2 Cryopreservation of gametes 8.3 Fish genetic resource and conservation 9. Omics approach in fisheries and aquaculture 10. Aquaculture economics in India 10.1 Commerce and trading in aquaculture 10.2 Socio-economic condition of fish farmers in India			12
Texts / References	1. Khanna and Singh, Textbook of Fish Biology and Fisheries (2015, Narendra Publishing House) 2. Gopakumar, Singh and Chitranshi, Fifty Years of Fisheries Research in India (2000, Fisheries Division, Indian Council of Agricultural Research) 3. Jhingran, Fish and Fisheries of India (1985, Hindustan Publishing Corporation) 4. Pillay, Aquaculture: Principles and Practices: Fishing News Books (2005, First Indian reprint) 5. Smith, Fish Diseases and Medicine (2019, CRC Press) 6. Mohanty, Advances in Fish Research Vol VII (2019, Narendra Publishing House)			

Learning Outcomes (not more than 100 words)	1. Students will learn about different types of aquaculture systems. 2. The importance of physicochemical requirements in aquaculture will be understood. 3. They will become familiar with the nutritional needs of species and approaches to disease management. 4. The advantages of the omics approach in aquaculture will be explored. 5. Awareness of the current socio-economic conditions of fish farmers will be gained.			
Course Title	ZOOMV52: Lab. Exercises based on course ZOOMV51			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• To learn about commercially important aquatic species. • To gain knowledge of different types of culture methods and fishing gear. • To learn effective water quality management practices for aquaculture.			
Course Content	The course content is divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Survey of fish market and identification of locally available fishes of economic importance			7
II	Seasonal analysis of pond water by measuring the following physico-chemical properties: i. Dissolved CO ₂ content ii. O ₂ content			8
III	Study of fishing nets			3
IV	Determination of protein content in a fresh and preserved fish			7
IV	Visit to fish farm and hatchery			5
Texts / References	1.Khanna and Singh, Textbook of Fish Biology and Fisheries (2015, Narendra Publishing House) 2. Gopakumar, Singh and Chitranshi, Fifty Years of Fisheries Research in India (2000, Fisheries Division, Indian Council of Agricultural Research) 3. Jhingran, Fish and Fisheries of India (1985, Hindustan Publishing Corporation) 4. Pillay, Aquaculture: Principles and Practices: Fishing News Books (2005, First Indian reprint) 5. Smith, Fish Diseases and Medicine (2019, CRC Press) 6. Mohanty, Advances in Fish Research Vol VII (2019, Narendra Publishing House)			
Learning Outcomes (not more than 100 words)	Students will be able to understand 1. Commercially important cultivable finfish and shellfish species. 2. Arrangement of different types of ponds and design of an ideal fish farm. 3. The analysis of water quality of a culture pond			

Course Title	ZOOMJ61: Functional Anatomy of Chordates			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	• To comprehend the comparative structures among chordates together with a discussion of the function and evolution of different animal physiological systems. • To provide vertebrate morphology as an integrative discipline bringing together physiology, embryology, behavior, and ecology deploying methods of systematics and paleontology.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Origin of vertebrates 1.1 Fish 1.2 Amphibia 1.3 Reptilia 1.4 Aves 1.5 Mammalia			5
II	2. Integument and its derivatives 2.1 Structure of integument 2.2 Scales, feathers, hair, claws, nails, hoofs, horns, antlers, glands			3
	3. Endoskeleton 3.1 General plan of neurocranium and dermatocranium. 3.2 Jaw suspensorium 3.3 Vertebrae			3
III	4. Digestive system 4.1 Modifications in the alimentary canal in relation to feeding habits: length and surface area, internal folds and supplementary diverticulae, ruminants' stomach 4.2 Dentition in mammals			5
	5. Respiratory system 5.1 Aquatic respiration 5.1.1 Cutaneous 5.1.2 Branchial 5.2. Aerial respiration 5.2.1 Accessory respiratory organs in fish 5.2.2 Air-sacs in birds 5.2.3 Lungs			6
IV	6. Circulatory system 6.1 Aortic arches 6.2 Portal systems 6.3 Lymphatic system			5
	7. Nervous system 7.1Evolution of the brain (cerebral hemispheres and cerebellum) 7.2 Neuromast organs of lower vertebrates			5
V	8. Urinogenital system 8.1 Excretory system 8.1.1 Types and evolution of kidney tubules 8.1.2 Urinary duct and bladder 8.2. Reproductive system 8.2.1 General plan of gonads and urinogenital ducts 8.2.2 Types of uteri			7
Texts / References	1.Kardong: Vertebrates: Comparative Anatomy, Function, Evolution (Recent edition, Mcgraw Hill Education) 2. Kent: Comparative Anatomy of the Vertebrates (Recent edition, Mcgraw Hill Education) 3. Romer & Parsons: The vertebrate Body (6 th ed. 1986, Saunders) 4. Jordan and Verma: Chordate Zoology (S. Chand & Company Pvt. Ltd.)			

Learning Outcomes (not more than 100 words)	1. The students can directly engage in the discovery of vertebrate form which includes various controversies in concepts that support them in understanding the scientific process. 2. A comprehensive comparative study of morphology, function, and evolution will help students to develop skills in critical thinking and mastery of concepts.
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Course Title	ZoomJ62: Environmental Biology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	• To understand about the living environment and the relationship between biological organisms and their environment • To study the interactions between humans and the natural environment and address the various environmental challenges, such as climate change, deforestation, air and water pollution, and biodiversity loss			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			

Units	Course Content	Hr. of Teaching
I	1. Introduction to environmental biology 1.1 Major ecosystems of the world-<i>tundra</i>, <i>taiga</i>, temperate deciduous forest, tropical rain forest, <i>savannah</i>, grassland and desert with adaptations in flora and fauna 2. Trophic relationship 2.1. Ecological pyramids 2.2. Food chain and Food web	10
II	3. Ecosystem energetics 3.1. Energy flow in ecosystem 3.2. Productivity 4. Population and communities 4.1. Population characteristics: density, natality, mortality and growth curve 4.2. Nature, structure and attributes of biological communities 4.3. Community dynamics: ecological succession, ecotone, edge effect 4.4. Positive and negative interactions	15
III	5. Global environmental changes 5.1. Sources and impact of environmental pollution – air, water and soil 5.2. Green house gases and their effects 5.3. Solid waste management 6. Conservation biology 6.1. Biodiversity: benefits, hotspots, threats and conservation 6.2. Soil and water resources and their conservation 6.3. Introduction to environmental laws and ethics	20

Texts / References	1. Reece et al. (2011): Campbell biology (9thed., Pearson, New York) 2. Cunningham and Cunningham (2003): Environmental science (Tata McGraw Hill Publishing Company Limited, New Delhi.) 3. Enger and Smith (2000): Environmental Science (7th ed., McGraw-Hill Higher Education, London) 4. Odum and Barrett (2005): Fundamental of Ecology, Thomson Brooks/Cole, USA 5. Primack (2004): A Primer of Conservation Biology (3rd ed., Sinauer Associates, USA) 6. J. S. Singh (2014): Ecology, Environmental Science and Conservation 7. P. D. Sharma (2007): Ecology and Environment
Learning Outcomes (not more than 100 words)	1. Studying environment biology not only enables students to learn a great deal of environment-related basic knowledge but also can foster a feeling toward the environment and develop skills needed to protect the environment 2. Environmental studies will provide you with the scientific knowledge and tools to measure, monitor, and model the environmental systems and processes, and to assess and mitigate the environmental impacts and risks of human activities and systems.

Course Title	ZOOMJ63: Cell Biology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hours of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- The course aims to provide an in-depth understanding of the organization of eukaryotic cells, principles of membrane transport, functions of different types of transporter proteins, and detailed processes associated with its functions like membrane transport, intracellular transport, signaling mechanisms, and cell reproduction. - Students will learn about the various cellular organizations, mitochondrial organization, cellular bioenergetics, and the structural organization of the cytoskeleton.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content	Hr. of Teaching		
I	- Eukaryotic Cell 1.1 Membrane Transport 1.1.1 Principles of membrane transport 1.1.2 Types of carrier proteins, channel proteins, and their regulation 1.1.3 Physiological importance of symporters and antiporters (Glucose transport, pH maintenance) 1.1.4 Types of transporters: P-type, F-type, V-type, ABC transporters 1.1.5 Gated channels 1.1.6 Selectively across Ion channels: K⁺, Na⁺ channels and Aquaporins 1.1.7 Endocytic-exocytic cycle 1.2 Intracellular transport and protein sorting 1.2.1 Types of signal sequences and their role in protein targeting 1.2.2 Passage of proteins through endoplasmic reticulum and membrane topology 1.2.3 Post-translational processing of protein in Endoplasmic reticulum and its significance 1.2.3 Processing and sorting of proteins in Golgi apparatus 1.2.4 Nuclear transport 1.2.5 Mitochondrial transport: TIM23 complex, SAM complex	7	8	
II	2. Mitochondria and energy transfer 2.1 Organization of mitochondrial genome 2.2 Fission and Fusion 2.3 Proton pumps of Electron transport chain 2.4 Electrochemical gradient and bioenergetics	4		
III	3. Cellular organization 3.1 Cytoskeleton 3.1.1 Structural organization of microfilaments and microtubules: Tread milling and Dynamic instability, actin-binding proteins and motor proteins. 3.1.2 Importance of Intermediate filaments: Keratin, Lamins, Desmin 3.1.3 Microtubular organelles 3.2 Cell-cell interactions 3.2.1 Types of Cell junctions 3.2.2 Properties of Cell adhesion and extracellular matrix 3.3 Cell signaling 3.3.1 Receptors: G-coupled receptors, RTK receptors, Nuclear receptors 3.3.2 Second messenger: cAMP and protein kinases 3.3.3 Overview of major signaling pathways	5	3	5

IV	4. Cell Reproduction 4.1 Cell Division 4.1.1 Experimental evidence of cell cycle events 4.1.2 M-phase: Biorientation, Chromosome movement, Regulation of Mitosis and Cytokinesis 4.1.3 Oncogenes and tumor suppressor genes 4.1.4 Cell transformation: Viral Transformation, Clonal expansion, angiogenesis and Metastasis 4.2 Apoptosis and cell death 4.2.1 Intrinsic and Extrinsic pathway 4.2.2 Bcl2 family	8 3
V	5. Interphase nucleus 5.1 Organization of interphase nucleus: Chromosome domains and territories, 3-C 5.2 Heterochromatin: HP1, Concept of boundary elements 5.3 Histones modifications and associated gene regulation 5.4 Centromere and telomere	5
Recommended Books	1. Alberts et al, Molecular Biology of the Cell (7th Edition, Garland) 2. Karp: Cell and Molecular Biology (9th Edition, John Wiley) 3. Lodish et al: Molecular Cell Biology (9th Edition, Freeman) 4. Plopper et al, Lewin's Cells (3rd Edition, Jones and Bartlett)	
Learning Outcomes (not more than 100 words)	1. Upon completing the course, students will have a comprehensive understanding of eukaryotic cells, detailed structures, and the various biological processes associated with eukaryotic cells. 2. Students will be proficient in analyzing and explaining the various processes involved in these cellular functions. 3. Students will be equipped with the necessary skills and insights to comprehend the complexities of eukaryotic cell biology and its significance in the broader field of life sciences.	

Course Title	ZOOMJ65: Lab. exercises based on course ZOOMJ61 & ZOOMJ62			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	<i>Functional Anatomy of Chordates</i> - To understand the comparative structures among chordates together with a discussion of function and evolution of different animal physiological systems through dissections, models and museum specimens. - To develop skills of dissection and basic laboratory work. <i>Environmental Biology</i> - To understand, analyze, and solve environmental problems using scientific methods and ethical principles			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content	Hr. of Teaching		
I	<i>PART-A: Functional Anatomy of Chordates (Credit 1)</i> 1. Lower chordates 1.1. Amphioxus 1.1.1 External features 1.1.2 Mounting of the oral hood, velum and pharyngeal wall 1.1.3 Study of the following slides: T.S. through oral hood, midgut diverticulum, pharyngeal region, gonads and post-oral region of the intestine 1.2 Study of the following permanent slides/specimens: <i>Pyrosoma</i>, <i>Salpa</i>, <i>Doliolum</i> 2. Vertebrates 2.1 Integument and its derivatives 2.1.1 Mounting of cycloid and ctenoid scales 2.1.2 Mounting of chromatophores of fish 2.1.3 Study of different types of feathers: Contour, filoplume and down feathers 2.2 General anatomy 2.2.1 Vascular system 2.2.1.1 Heart and afferent and efferent branchial vessels of <i>Mystus/Cirrhinus sp.</i> 2.2.1.2 Arterial and venous systems of rat 2.2.2 Respiratory system: Accessory respiratory organs of <i>Heteropneustes/ Channa/Clarius</i>	15		

II	3. Histology: Study of permanent slides of the following tissues and organs of mammals- Tooth, tongue, oesophagus, stomach, intestine, pancreas, liver, spleen, kidney, cartilage, bone 4. Classification and distinctive features of the following 4.1 Cyclostomata: <i>Petromyzon</i>, <i>Myxine</i> 4.2 Chondrichthyes: <i>Scoliodon</i>, <i>Sphyrna</i>, <i>Torpedo</i>, <i>Pristis</i> 4.3 Osteichthyes: <i>Protopterus</i>, <i>Exocoetus</i>, <i>Hippocampus</i>, <i>Syngnathus</i>, <i>Tetradon</i>, <i>Diodon</i>, <i>Amia</i>, <i>Anabas</i>, <i>Ophiocephalus</i>, <i>Clarias</i>, <i>Heteropneustes</i>, <i>Catla</i>, <i>Labeo</i>, 4.4 Amphibia: <i>Ichthyophis</i>, <i>Axolotl</i> larva, <i>Amphiuma</i>, <i>Proteus</i>, <i>Pipa</i>, <i>Rhacophora</i>, <i>Hyla</i> 4.5 Reptilia: <i>Turtle</i>, <i>Chameleon</i>, <i>Draco</i>, <i>Uromastix</i>, <i>Varanus</i>, <i>Calotes</i>, <i>Iguana</i>, <i>Mabuya</i>, <i>Xenotrix</i>, <i>Naja</i>, <i>Vipera</i>, <i>Alligator</i> 4.6 Aves: <i>Apteryx</i>, <i>Struthio</i>, <i>Aptenodytes</i>, <i>Francolinus</i>, <i>Tyto alba</i>, <i>Dinopium</i>, <i>Milvus</i>, <i>Corvus</i>, <i>Pavo</i>, <i>Eudynamis</i>, <i>Passer</i>, <i>Psittacula</i>, <i>Anas</i>, <i>Grus</i> 4.7 Mammalia: <i>Ornithorhynchus</i>, <i>Tachyglossus</i>, <i>Macropus</i>, <i>Manis</i>, <i>Erinaceus</i>, <i>Pteropus</i>, <i>Lemur</i>, <i>Loris</i>, <i>Bradypus</i>, <i>Phoca</i>, <i>Lutra</i>, <i>Equus caballus</i>, <i>Camelus</i>, <i>Capra</i>, <i>Bos</i>	15
III	<i>PART-B: Environmental Biology (Credit 1)</i> 1. Estimation of grasshopper population density by capture-recapture method 2. Estimation of percent frequency, abundance and dominance of producers in grassland by quadrat sampling method 3. Estimation of percent frequency, abundance and dominance of consumers in grassland by quadrat sampling method	15
IV	4. Determination of total hardness of water sample 5. Determination of pH of different water samples by using pH meter. 6. Determination of carbonate and nitrate in soil samples 7. Determination of free CO₂ in different samples of pond water	15
Texts / References	<i>Functional Anatomy of Chordates</i> 1. Lal: Practical Zoology- Vertebrates (12th ed., Rastogi Publications) 2. Verma: A manual of Practical Zoology-Chordates (11th ed., S Chand publications) <i>Environmental Biology</i> 1. Reece et al. (2011): Campbell biology (9thed., Pearson, New York) 2. Cunningham and Cunningham (2003): Environmental science (Tata McGraw Hill Publishing Company Limited, New Delhi.) 3. Enger and Smith (2000): Environmental Science (7th ed., McGraw-Hill Higher Education, London) 4. Odum and Barrett (2005): Fundamental of Ecology, Thomson Brooks/Cole, USA 5. Primack (2004): A Primer of Conservation Biology (3rd ed., Sinauer Associates, USA) 6. J. S. Singh (2014): Ecology, Environmental Science and Conservation 7. P. D. Sharma (2007): Ecology and Environment	
Learning Outcomes (not more than 100 words)	1. Students will gain familiarity with the anatomy of representative animals. 2. They will understand organ function, evolution, and the placement of organisms within a phylogenetic context. 3. Students will acquire scientific knowledge and tools to measure, monitor, and model environmental systems, assess impacts, and mitigate the risks of human activities.	

Course Title	ZoomJ66: Lab. exercises based on course ZoomJ63 & ZoomJ64			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits-Section B		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	- Equip students with hands-on experience in advanced microscopy techniques, including phase contrast, fluorescence, confocal, and electron microscopy. - Develop proficiency in cell biology practices, such as tissue culture, chromosome analysis, and DNA visualization techniques. - Enable students to perform essential mammalian physiology procedures, including blood cell counting, hemoglobin determination, and blood pressure measurement.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	PART-A: Cell Biology (Credit 1) - Use of phase contrast optics by observing the live sperm/paramecium in physiological saline - Demonstration of the use of fluorescence microscope (DNA and RNA fluorescence) and familiarity with confocal microscope - Familiarity with scanning and transmission electron microscopes (photograph and visit the Department of Metallurgy) - Familiarity with tissue culture (visit plant and animal tissue culture labs.) Use of colchicine in arresting anaphase movement (onion root tips) - Visualization heterochromatin zed sex chromatin through Barr body - Demonstration of chromatin distribution using DAPI staining. - Metaphase chromosome preparations from the bone marrow of mouse/rat - Meiosis in grass hopper testis - Study of permanent slides to understand the following: - Constitutive heterochromatin(C-band) - Semiconservative DNA replication (BrdU-labelling) - Detection of S-phase cells in a random population (autoradiography) - Transcription in polytene chromosome puffs (autoradiography) - Chromosome organization (G-banding) - Premature chromosome condensation - Actin remodeling in migrating tumor cells - Prediction of telomere length through FISH.			30
II	PART-B: Mammalian Physiology (Credit 1) - Counting of red blood corpuscles - Counting of white blood corpuscles - Determination of hemoglobin content - Determination of blood groups - Measurement of blood pressure using sphygmomanometer - Preparation of casein from milk			30
Texts / References				
Learning Outcomes (not more than 100 words)	- Students will develop the ability to analyze cellular processes through advanced microscopy and experimental techniques, enhancing their understanding of chromosomal and molecular biology. - They will gain hands-on experience in essential physiological procedures, enabling them to accurately assess and interpret key mammalian biological functions.			

Course Title	ZOOMV61: Clinical Biochemistry			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- Equip students with an understanding of the biochemical basis of various diseases, including enzyme and protein deficiency disorders, and related diagnostic and treatment strategies. - Develop knowledge of the chemistry, composition, and functions of bodily fluids, along with common biomarkers of organ functions. - Provide insights into the mechanisms of drug action, including biotransformation, drug delivery, and the impact of addiction.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Biochemical basis of diseases/disorders, diagnosis and treatment 1.1. Molecular deficiency disorders 1.1.1 Disorders of enzyme deficiency 1.1.2 Disorders of protein deficiency/defects 1.2. Storage and transport associated disorders Biochemistry of Diabetes and Obesity 1.3. Biochemical changes in pregnancy & lactation			10
II	2. Chemistry, composition & functions 2.1 Blood 2.2 Lymph 2.3 CSF 2.4 Ascitic fluid 2.5 Pleural fluid 2.6. Lipid profile			10
III	3. Common biomarkers of organ functions 3.1 Liver 3.2 Kidney 3.3 Thyroid 3.4 Heart			10
IV	4. Immunity and human health 4.1 Autoimmunity 4.2 Immunodeficiency 4.3 Cancer 4.4 Septicemia 4.5 Transplantation			8
V	5. Drugaction 5.1 Mechanisms 5.2 Drug addiction 5.3 Biotransformation of drugs 5.4 Drug delivery, targeting and clinical trials			7
Texts / References	1. Bhagvan: Medical Biochemistry (4th Ed.), Hap, 2004 2. Goodman et al.: The Pharmacological Basis of Therapeutics (7th Ed.), Macmillan Publishers, 1991			
Learning Outcomes (not more than 100 words)	Students will develop the ability to i. Learn how biochemical imbalances and abnormalities lead to various diseases. ii. Interpret laboratory results and make informed clinical decisions. iii. A comprehensive understanding of the biochemical basis of human health and disease. iv. Apply biochemical principles to interpret laboratory test results. v. Perform common clinical biochemistry laboratory techniques and procedures. vi. Analyze and interpret biochemical data to make informed clinical decisions.			

Course Title	ZOOMV62: Lab. Exercises based on course ZOOMV61			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• Train students in the proper techniques for specimen collection, handling, and storage to ensure accurate biochemical analysis. • Develop proficiency in the estimation of key enzyme activities, including LDH, AST, ALT, and alkaline phosphatase. • Equip students with the skills to measure and interpret serum lipid profiles, glucose, and urea levels.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Specimen collection, handling & storage of sample. 2. Estimation of total activity of following enzymes. i) LDH ii) AST (GOT) iii) ALT (GPT) iv) Alkaline phosphatase 3. Estimation of serum lipid profile i) Serum total cholesterol. ii) Serum HDL cholesterol. 4. Estimations i) Glucose ii) Urea			30
Texts / References	1. Bhagvan: Medical Biochemistry (4th Ed.), Hap, 2004 2. Goodman et al.: The Pharmacological Basis of Therapeutics (7th Ed.), Macmillan Publishers, 1991			
Learning Outcomes (not more than 100 words)	• Students will be able to apply best practices in laboratory settings for the preservation of sample integrity during collection and storage. • They will accurately perform and analyze enzyme assays, demonstrating competence in quantifying enzyme activities in clinical samples. • Students will interpret lipid profile data, identifying potential health risks associated with abnormal cholesterol levels. • They will develop critical thinking skills by analyzing glucose and urea levels, drawing correlations to metabolic and renal health.			
Course Title	ZOOMJ71: Biochemistry and Molecular Biology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	• The course covers various fundamental and advanced topics in genetics. • The course aims to provide students with a comprehensive understanding of the principles and mechanisms of genetics. • Students will learn how genes are mapped to chromosomes and study various tools to understand their regulation and function. • This course will also assist towards understanding of biological processes like transposition, genetic determination of sex, DNA recombination, damage and repair. • The course starts by exploring genetics in bacteria and progresses to the mammalian genome.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			

Units	Course Content	Hr. of Teaching	
I	1. Proteins: Composition and organization 1.1 Amino acids: Ionization, titration curve, pK and pI 1.2 N-terminal analysis: Sanger and Edman's reactions 1.3 Higher order organization 2. Enzymes 2.1 Kinetics (determination of Km and Vmax using Michaelis-Menten and Lineweaver-Burk plots) 2.2 Acid-base and covalent catalysis 2.3 Concept of regulation of enzyme activity 2.3.1 Inhibition 2.3.2 Allosteric regulation 2.3.3 Role of covalent modifications 2.4 Ribozymes and concept of abzymes	5 7	
II	3. Carbohydrates 3.1. Polysaccharides 3.1.1 Homopolymers 3.1.2 Heteropolymers (peptidoglycans and glycosaminoglycans) 3.2. Catabolism of carbohydrates and ATP production 3.2.1 Glycolysis 3.2.2 Krebs cycle 3.2.3 Electron transport chain and ATP synthesis 4. Lipids: Structural and functional significance 4.1 Triglycerides 4.2 Phospholipids 4.3 Sphingolipids 4.4 Cholesterol	7 3	
III	5. Nucleic acids 5.1 Conformation of DNA (A, B and Z) 5.2 Hoogsteen base pairing-Guanine quadruplex 5.3 Structure of nucleosomes 5.4 Mechanism of DNA replication 5.5 Transcription 5.5.1 Basic mechanism (prokaryotic model) 5.5.2 Transcription initiation complex in eukaryotes 5.5.3 Processing of RNA Pol II Transcript 5.5.4 Mechanism of translation	12	
IV	6. Recombinant DNA and Proteins 6.1 Tools: Restriction enzymes, vectors 6.2 Construction of recombinant DNA 6.3 Production of recombinant protein 6.4 Genome editing	4	
Texts / References	1. Lehninger, Nelson & Cox: Principles of Biochemistry (4 th ed, 2007, Worth), 2. Berg, Tymoczko, Stryer; Biochemistry (7 th ed. 2012, Freeman)		
Learning Outcomes (not more than 100 words)	1. Students will have a sound understanding of the biochemical processes that underpin biological functions at a molecular level and effectively communicate scientific findings relevant to biomolecules ensuring clarity and accuracy. 2. Students will develop the capability to understand biomolecule types, analyze structure-function relationships, interpret biological interactions, describe metabolic pathways, apply laboratory techniques, solve biochemical problems, and communicate scientific ideas.		

Course Title	ZOOMJ72: Molecular Genetics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- The course covers various fundamental and advanced topics in genetics. - The course aims to provide students with a comprehensive understanding of the principles and mechanisms of genetics. - Students will learn how genes are mapped to chromosomes and study various tools to understand their regulation and function. - This course will also assist towards understanding of biological processes like transposition, genetic determination of sex, DNA recombination, damage and repair. - The course starts by exploring genetics in bacteria and progresses to the mammalian genome.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content	Hr. of Teaching		
I	1. Gene mapping 1.1 Conjugation, transduction and transformation in Bacteria 1.2 Tetrad Analysis in <i>Neurospora</i> 1.3 Three-point test cross in <i>Drosophila</i> 1.4 Introduction to physical mapping in human 2. Gene expression and gene regulation 2.1 Inducible (<i>lac</i>) and repressible (<i>trp</i>) operons in <i>E. coli</i> 2.2 Regulation of gene activity in eukaryotes 2.2.1 Transcription unit 2.2.2 Initiation of transcription 2.2.3 Transcription factors: repressors, activators 2.2.4 Enhancers and silencers 2.2.5 Concept of epigenetic regulation	6		
II	3. Transposable elements 3.1 Discovery of transposable elements 3.2 Prokaryotic and eukaryotic elements 3.3 Retroposons 3.4 Mechanisms of transposition 4. Organization of the Human Genome 4.1 Human Genome Project 4.2 Nuclear and Mitochondrial Genome 4.3 Gene families 4.4 LINE& SINE elements 4.5 Single Nucleotide Polymorphism 5. Genetic determination of sex 5.1 <i>Drosophila</i> 5.2 Human	6		
III	6. DNA recombination 6.1 Homologous and non-homologous recombination 6.2 Site-specific recombination 7. DNA damage and repair 7.1 Endogenous and exogenous origin of DNA damage 7.2 Major repair pathways: BER, NER, mismatch repair, NEJ and DSB 8. Human genetic disorders 8.1 Metabolic: Phenylketonuria 8.2 Triplet repeat expansion: Huntington 8.3 Multifactorial: Diabetes mellitus	4		
8			8	
3			3	
Texts / References	1. Snustad and Simmons: Principles of Genetics (2019, 7th ed. John Wiley) 2. Pierce: Genetics - A conceptual approach (2012, Freeman) 3. Watson et al.: Molecular biology of the gene (7th ed 2017)			

	4. Gardner et al: Principles of Genetics (2006, John Wiley) 5. Griffith et al: An Introduction to Genetic Analysis (Freeman, 2008) 6. Russell: Genetics (2010, Benjamin Cummings) 7. Genome: The Autobiography of a species in 23 chapters: Matt Ridley 2017. Harper Collins
Learning Outcomes (not more than 100 words)	1. Upon successfully completing the ZOB 603 Genetics course, students can understand how the chromosome maps are prepared. 2. They will also learn about the significance of mutations, aiding their understanding of gene regulation and function. 3. This course will provide a foundational understanding of genetics, which is essential for grasping advanced genetic concepts, and will open up several avenues for students in terms of research and employability.

Course Title	ZOOMJ73: Molecular Endocrinology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- To explain the mechanisms of action of protein and steroid hormones, including signaling pathways and genomic/nongenomic effects. - To understand the organization and functions of the hypothalamo-hypophyseal system, including the roles of hypophysiotropic, adenohypophyseal, and neurohypophyseal hormones. - To explore the epiphysial system, focusing on the pineal gland, biological rhythms, and melatonin's role in photoperiodic measurement. - To describe steroid hormone biosynthesis pathways. - To examine the structure and physiological roles of androgens, estrogen, progesterone, inhibin, mineralocorticoids, glucocorticoids, and catecholamines. - To analyze the regulation and functions of parathormone, calcitonin, vitamin D, insulin, and glucagon in homeostasis.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content	Hr. of Teaching		
I	Mechanism of hormone action 1.1 Protein hormones: Membrane receptors and signaling pathways (G-proteins, cyclic AMP Signaling cascade, PKC and CaMK signaling pathway), receptor tyrosine kinase pathway 1.2 Steroid hormones: Genomic and nongenomic pathways	10		
II	2.1 Hypothalamo-hypophyseal System 2.1.1 General organization 2.1.2 Neural control of adenohypophysis: hypophysiotropic hormones and their actions 2.1.3 Adenohypophyseal hormones & their physiological roles - Somatotropin and prolactin - Glycoprotein hormones: FSH, LH and TSH - Pro-opiomelanocortin: ACTH, MSH 2.1.4 Neurohypophyseal hormones: oxytocin and vasopressin 2.2 Epiphysial System 2.2.1 Pineal, biological clock and calendar 2.2.2 Melatonin and photoperiodic measurement	15		
III	Hormonal control of 3.1 Calcium homeostasis 3.2 Glucose homeostasis	6		
IV	Adrenal cortex: Control of secretion and physiological role of mineralocorticoid and glucocorticoid hormone Adrenal medulla: Catecholamine biosynthesis, release and physiological role	5		
V	Biology of Gonads:	9		

	5.1 Gonadal development 5.2 Physiology of the male and female reproductive axis 5.3 Sexual dysfunctions			
Texts / References	1. Hadley: Endocrinology, Fifth Edition, 2000, Prentice Hall 2. Norris: Vertebrate Endocrinology, Fifth Edition, 20012, Academic Press 3. Goodman: Basic Medical Endocrinology, Fifth Edition, 2016, Elsevier Inc. 4. Larson: Williams Textbook of Endocrinology (15 th ed 2022, Saunders)			
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to: 1. Understand signalling mechanisms of protein and steroid hormones. 2. Describe the structure and function of the hypothalamo-hypophysial system. 3. Explain the roles of adenohypophysial hormones. 4. Understand neurohypophysial hormones. 5. Explain the function of the pineal gland and melatonin. 6. Describe steroid hormone biosynthesis. 7. Understand the roles of androgens, estrogens, progesterone, inhibin in gonadal function. 8. Describe the secretion and function of adrenal cortex hormones (mineralocorticoids, glucocorticoids) and adrenal medulla catecholamines. 9. Explain calcium and glucose homeostasis involving parathormone, calcitonin, vitamin D, insulin, and glucagon.			
Course Title	ZOOMJ74: Bioinformatics and Biostatistics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above):			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	Bioinformatics: - To provide detailed information about various tools and methodologies to analyze large datasets quickly and efficiently. - To identify pattern and relationship in the data that would be difficult to identify manually - To predict novel targets for developing safe and effective drugs against complex and complicated animal diseases. Biostatistics: - To provide essential tools and methodologies for analyzing data derived from biological, medical, and health-related fields. - To identify health trends, assessing risk factors, and evaluating the effectiveness of treatments and interventions. This information is vital for public health planning and policy-making, aiming to improve population health outcomes.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Bioinformatics: 1. Introduction of bioinformatics and basics of information technology 1.1 Operating system, web browser, internet protocol (TCP/IP), hypertext, HTML, Web page and uniform resource locators (URL) 2. Introduction to data archiving systems 2.1 FASTA format 2.2 Accession, and GI-Number			4
				2

II	3. Basic features of biological data and data 3.1 Types of Biological Data- Primary Databases, Secondary and Composite Databases 3.2 Nucleic acid sequences databases (GenBank, EMBL, DDBJ) 3.3 Genome databases 3.4 Protein sequence, structures and interacting protein databases (Swiss-Prot, TrEMBL, UniProt, PDB) 3.5 Literature databases	6
	4. Introduction to data retrieval systems 4.1 Search engines 4.2 Entrez, sequence retrieval system (SRS) and DBGET / LinkDB	2
III	5. Introduction to molecular sequence analysis software packages and tools 5.1 Prediction of motifs and domains (Prosite, ProDom, Pfam, InterPro) 5.2 Sequence (BLAST and Clustal W) and phylogenetic analysis	4
	6. Scope and applications of bioinformatics	2
IV	Biostatistics: 1.1. Data collection and classification 1.2. Graphical representation of data 1.1.1. Pie chart 1.1.2. Bar diagram 1.1.3. Histogram 1.1.4. Frequency polygon 1.1.5. cumulative frequency curve	06
V	2.1. Basics of Probability 2.2. Measures of central tendency: Arithmetic Mean, Median, Mode 2.3. Measures of dispersion: Standard deviation and Standard error, Concept of Coefficient of Variation 2.4. Correlation: Types of correlation, Calculation of correlation in continuous data and ordinal data 2.5. Regression: Linear regression and regression equations 2.6. Hypothesis testing: Parametric tests (Paired and unpaired t-test, ANOVA), Non-parametric test (Chi-square test, Mann-Whitney U test)	13
Texts / References	Bioinformatics 1. Barnes & Gray: Bioinformatics for geneticists (2003, Wiley) 2. Lesk: Bioinformatics (5th ed 2015, Oxford) 3. Westhead et al: Bioinformatics Instant Notes (Indian ed 2003, Viva Books) 4. Mount, Bioinformatics (2 nd ed 2006, CBS) 5. Hunt and Livesey: Functional Genomics (2006, Oxford) 6. Campbell: Discovering Genomics, Proteomics and Bioinformatics (2006, LPE) 7. A textbook of bioinformatics, Sharma, Munjal and Shanker (2016, Rastogi Publications) 8. Hamid D. Ismail (2022, Chapman and Hall/CRC) Bioinformatics: A Practical Guide to NCBI Databases and Sequence Alignments Biostatistics: 1. Bruning & Kintz (1977) Computational Handbook of Statistics. Scott 2. Daniel (2000) Biostatistics: A Foundation for Analysis in Health Sciences. John Wiley. 3. Rastogi (2008) Fundamentals of Biostatistics. ANE Books 4. Sharma (2008) Text Book of Biostatistics-I & II. Discovery Publishing 5. Zar (2003) Biostatistical Analysis. Pearson	
Learning Outcomes (not more than 100 words)	1. After completing the course, students will develop basic computer skills and become proficient in using genomic, proteomic, and literature databases. 2. They will learn to compare and analyze biological sequences, construct phylogenetic trees, and identify consensus sequences, genes, and open reading frames. 3. Students will also gain the ability to identify protein motifs and domains, and design therapeutic drugs targeting novel disease mechanisms. 4. Students will gain proficiency in collecting, representing, and analyzing research and health-related data, and will be able to effectively communicate these findings to support decision-making in clinical, public health, and research contexts.	

Course Title	ZOOMJ75: Lab. exercises based on course ZOOMJ71& ZOOMJ72			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	- The practical course is designed to give students hands-on training in genetics-based studies. - This will be accomplished through a series of laboratory exercises, including mutant analysis and simulation-based experiments to demonstrate inheritance principles. - Students will engage in various hands-on exercises to better grasp concepts such as sex-linked inheritance and the preparation and analysis of different types of chromosomes.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	PART-A: Structure and Functions of Biomolecules (Credit 1) - Determination of pK and pI values of glycine - Estimation of protein by Biuret method - Determination of the activity of enzyme (Urease): - Effect of [S] and determination of Km and Vmax - Effect of time - Qualitative tests for identification of sugars - Qualitative test for DNA & RNA			30
II	PART-B: Molecular Genetics (Credit 1) - Study of mutants in <i>Drosophila</i> (Bar eye, white eye, yellow body, sepia eye, curled wing, dumpy wing, vestigial wing and sepia eye-curled wing and curled wing-ebony body- sepia eye). - Simulation of principles of segregation and independent assortment using coloured beads. - Application of laws of probability and chi-square test. - Study of sex-linkage: reciprocal crosses between red and white-eyed <i>Drosophila</i> flies. - Study of pattern of inheritance of the traits in human populations: rolling of tongue and interlocking, and of the sex-influenced trait long versus short second finger in relation to the fourth finger (apply Hardy-Weinberg Law). - Preparation of polytene chromosome from <i>Drosophila</i> larvae. - Isolation of Genomic DNA from mouse/rat tissue. - Population & Familial analysis of Diabetes.			30
Texts / References				
Learning Outcomes (not more than 100 words)	- The hands-on Molecular Genetics course offers students a unique opportunity to gain practical skills. - By observing genetic mutations in <i>Drosophila</i>, students can correlate phenotype with genotype, understand genetic interaction and their molecular basis. - Students will be able to set hands on genetic crosses to understand recessive and dominant, segregation, pattern of inheritance and finally evaluating statistical significance. - Students will acquire comprehensive experience in gathering data from human populations and exploring the intricacies of sex-linked inheritance. - They will also know how to prepare and examine chromosomes for various downstream applications, which will deepen their understanding of the genetic basis of inheritance.			

Course Title	ZOOMJ76: Lab. exercises based on course ZOOMJ73 & ZOOMJ74			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	<i>Molecular Endocrinology:</i> - Students will develop skills in handling and maintaining laboratory rats, survey endocrine glands and reproductive organs, and perform surgical ablation of testis and ovary. - They will study the estrous cycle through vaginal smear analysis, prepare permanent histological slides of endocrine glands, and use ELISA for quantitative hormone assays. <i>Bioinformatics:</i> - Students will learn to use and develop tools for curating, comparing, and analyzing biological data, and identify patterns and relationships that are challenging to detect manually. <i>Biostatistics:</i> - Students will understand various tools and techniques for analyzing data in biology, medicine, and health contexts.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content	Hr. of Teaching		
I	<i>Molecular Endocrinology(Credit 1)</i> 1.1 Handling, sexing and maintenance of laboratory rat 1.2 Surgical ablation of testis and ovary in rat 1.3 Study of the oestrous cycle of rats by vaginal smear preparation	12		
II	2.1 Survey of endocrine glands and reproductive organs in rat 2.2 Preparation of permanent slides of endocrine glands: - Pituitary - Thyroid - Testis - Ovary - Pancreas - Uterus 2.3 Immunohistochemical localization of hormone receptors in reproductive tissues (Demonstration only)	18		
III	<i>Bioinformatics: (Credit 0.5)</i> 1. Familiarization with computer operations and TCP/IP 2. Data archiving systems: FASTA format, Accession and GI numbers 3. Use of search and meta-search engines (Google, Altavista, Dogpile)	15		
III	4. Demonstration of web pages related to biological information: - NCBI - ExPasy 5. Hands-on practice with features of the following databases: - GenBank - PDB - PubMed 6. Hands-on practice on following software packages/tools: - BLAST, - Clustal-W - Prosite - NetPhos - Rasmol			
IV	<i>Biostatistics (Credit 0.5)</i> 1.1. Presentation of data in the form of pie chart, bar diagrams, Histogram, frequency polygon and cumulative frequency curve. 1.2. Use of chi-square test to analyze biological data.	15		

	1.3. Calculation of student t-test (paired and unpaired both). 1.4. Applying one-way ANOVA to test differences among more than two	
Texts / References	1. Advanced Practical Zoology by PS Verma & PC Srivastava. S. Chand Publishing, 2015 2. Bruning & Kintz (1977) Computational Handbook of Statistics. Scott 3. Daniel (2000) Biostatistics: A Foundation for Analysis in Health Sciences. John Wiley. 4. Rastogi (2008) Fundamentals of Biostatistics. ANE Books 5. Sharma (2008) Text Book of Biostatistics-I&II. Discovery Publishing 6. Zar (2003) Biostatistical Analysis. Pearson	
Learning Outcomes (not more than 100 words)	<i>Molecular Endocrinology</i> Students will master the handling of laboratory rats, identify endocrine glands and reproductive organs, and perform testis and ovary ablation. They will analyze the estrous cycle through vaginal smears, prepare histological slides of endocrine organs, and conduct quantitative hormone assays using ELISA. <i>Bioinformatics</i> Students will gain skills in computational interpretation of biological data, and analyze biological sequences. They will locate consensus sequences, genes, and open reading frames, and infer evolutionary relationships to identify novel drug targets. <i>Biostatistics:</i> Students will practice analyzing collected and experimental data, select and apply appropriate parametric or nonparametric tests, and interpret results for decision-making.	

Course Title	ZOOMN71: Structure and Functions of Biomolecules			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- The main objective of Structure and Functions of Biomolecules course is to understand the fundamental concepts of biomolecules, including the definition, classification and organization of proteins, enzymes, carbohydrates, lipids, and nucleic acids. - This course emphasized the structural properties of major biomolecules and explain how these structures relate to their functions in biological systems that guide students through the complexities of biomolecular science.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content	Hr. of Teaching		
I	Proteins and Enzymes: Composition and organization 1.1 Amino acids: Ionization, titration curve, pK and pI 1.2 N-terminal analysis: Sanger and Edman's reactions 1.3 Higher order organization of proteins 1.4 Kinetics (determination of Km and Vmax using Michaelis-Menten and Lineweaver-Burk plots) 1.5 Acid-base and covalent catalysis 1.6 Concept of regulation of enzyme activity 1.6.1 Inhibition 1.6.2 Allosteric regulation 1.6.4 Role of covalent modifications 1.7 Ribozymes and concept of abzymes	15		
II	Carbohydrates 2.1. Polysaccharides 2.1.1 Homopolymers 2.1.2 Heteropolymers (peptidoglycans and glycosaminoglycans) 2.2. Catabolism of carbohydrates and ATP production 2.2.1 Glycolysis 2.2.2 Krebs cycle 2.2.3 Electron transport chain and ATP synthesis	8		
III	Lipids: Structural and functional significance	5		

	3.1 Triglycerides 3.2 Phospholipids 3.3 Sphingolipids 3.4 Cholesterol	
IV	Nucleic acids 4.1 Conformation of DNA (A, B and Z) 4.2 Hoogsteen base pairing-Guanine quadruplex 4.3 Structure of nucleosomes 4.4 Mechanism of DNA replication 4.5 Transcription 4.5.1 Basic mechanism (prokaryotic model) 4.5.2 Transcription initiation complex in eukaryotes 4.5.3 Processing of RNA Pol II Transcript 4.5.4. Mechanism of translation	12
V	Recombinant DNA and Proteins 5.1 Tools: Restriction enzymes, vectors 5.2 Construction of recombinant DNA 5.3 Production of recombinant protein 5.4 Genome editing	5
Texts / References	1. Lehninger, Nelson & Cox: Principles of Biochemistry (4 th ed, 2007, Worth), 2. Berg, Tymoczko, Stryer; Biochemistry (7 th ed. 2012, Freeman)	
Learning Outcomes (not more than 100 words)	1. Upon completion of the course students have a sound understanding of the biochemical processes that underpin biological functions at a molecular level and effectively communicate scientific findings relevant to biomolecules ensuring clarity and accuracy. 2. Students will develop the capability to understand biomolecule types, analyze structure-function relationships, interpret biological interactions, describe metabolic pathways, apply laboratory techniques, solve biochemical problems, and communicate scientific ideas.	

Course Title	ZOOMN72: Lab. exercises based on course ZOOMN71			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits-Section A		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	• Develop practical skills in handling biological materials and laboratory equipment. • Enhance understanding of the practical concepts related to the structure and function of biomolecules. • Develop critical thinking and problem-solving skills by evaluating experimental data and understanding the limitations of current biomolecular techniques. • Apply experimental methodologies to investigate the properties of biomolecules. • Develop effective communication skills through lab reports and presentations.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content-		Hr. of Teaching	
I	1. Determination of pK and pI values of glycine 2. Estimation of protein by Biuret method 3. Determination of the activity of enzyme (Urease): a. Effect of [S] and determination of Km and Vmax b. Effect of time 4. Qualitative tests for identification of sugars 5. Qualitative test for DNA & RNA		30	
Texts / References	1. Plummer and Plummer: Introduction to Pract. Biochem. (2001) Tata McGraw Hill 2. Wilson and Walker: Principles of Biochemical and Molecular Biological Techniques (6th ed2006, Cambridge University Press)			
Learning Outcomes	Students will be able to- 1. Collect, analyze, interpret, and present experimental data related to biomolecule structure and			

(not more than 100 words)	function. 2. Safely and accurately perform standard laboratory techniques related to biomolecule isolation, purification, and analysis. 3. Apply knowledge of biomolecules to drug development, biotechnology etc. 4. Design and conduct independent research projects related to biomolecules.
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Course Title	ZoomJ81: Biotechniques			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- To learn basics of instrumentation including principle and applications in the field of Biological Sciences. - Familiarization with cell culture techniques and methodology.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Analytical instruments: Principle and applications& Microscopy 1.1 pH meter 1.2 UV-visible spectrophotometer 1.3 Centrifuge 1.4 Bright field 1.5 Dark field 1.6 Polarized and differential interference contrast microscopy 1.7 Atomic force microscopy 1.8 Fluorescence and Confocal 1.9 Scanning and transmission electron microscopes			15
II	2. Histological Techniques 2.1 Fixation and Fixatives 2.2 Tissue processing and Microtomy 2.3 Principle and methods of histological staining 2.3.1 Hematoxylin and eosin 2.4 Principles of immunohistochemistry			7
III	3. Cell and tissue culture techniques 3.1 Sterilization: laminar flow, media and glassware 3.2 Types of animal cell culture 3.3 Cell viability testing 3.4 Cryopreservation			8
IV	4. Separation techniques 4.1 Chromatography: Principle and Applications 4.1.1 Paper chromatography 4.1.2 Thin layer chromatography 4.1.3 Ion exchange chromatography 4.1.4 Gel-filtration chromatography 4.1.5 HPLC 4.2 Electrophoresis: Principle and applications of 4.2.1 Agarose gel electrophoresis, native and SDS-PAGE 4.2.2 Western blotting			10
V	5. Radiotracer and Immunoassays techniques: Principle and Applications 5.1 Radioactivity: Nature, measurements and safety aspects 5.2 Immunoassay techniques 5.2.1 RIA 5.2.2 ELISA			5
Texts / References	1. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,) 2. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-			

	Living stones) 3. Bancroft: Theory and Practice of Histological Techniques (2018 Kim S. Suvarna, Christopher Layto, John D. Bancroft) 4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill) 5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)			
Learning Outcomes (not more than 100 words)	Students will be equipped to handle various instruments and learn techniques related to sampling, imaging, data collection, analysis and interpretation in biological samples			
Course Title	ZOOMJ82: Neurobiology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	The neurobiology course will be able to describe the structure and function of the brain at the sub-cellular and systems level of analysis and explain how molecular and electrical activity in the nervous system contributes to sensation, movement, cognition, and some diseases of the nervous system			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Organization of nervous system 1.1 Brain structure 1.2 Neurons and glia 1.3 Neuronal system: limbic and extrapyramidal 1.4 Blood brain barrier 1.5 Cerebrospinal fluid			10
II	Axonal and synaptic transmission 2.1 Membrane potential, action potential, transmission of nerve impulse 2.2 Types of synapses 2.3 Excitatory and inhibitory post-synaptic potential 2.4 Chemical transmission, neurotransmitters (acetylcholine, catecholamines, serotonin and GABA), neuropeptides			10
III	Learning, memory and Behaviour 3.1 Types and tools 3.2 Molecular mechanisms 3.3 Motivation 3.4 Sleep			10
IV	Brain aging 4.1 Structural alterations 4.2 Functional changes 4.3 Neurogenesis and neuroplasticity 4.4 Neurodegeneration and neuroprotection			5
V	Neuropathology and Brain Imaging 5.1 Strokes 5.2 Epilepsy 5.3 Alzheimer disease 5.4 Huntington disease 5.5 Parkinson disease Brain imaging 5.6 CAT 5.7 PET 5.8 MRI			10
Texts / References	1. Guyton and Hall: Text Book of Medical Physiology (11th Ed 2006, W.B. Saunders) 2. Eric Kandel: Principles of Neural Science (2000, Mc Graw Hill) 3. Allan Siegel (Author), Hreday N. Sapru (Author) Essential Neuroscience (4th Edition 2014 Lippincott Williams and Wilkins)			

	4. Mark F. Bear (Author), Barry W. Connors (Author), Michael A. Paradiso (Author)Neuroscience: Exploring the Brain (4 th Edition 2015 Wolters Kluwer)			
Learning Outcomes (not more than 100 words)	- Students will demonstrate proficiency of the structure and function of the nervous system at various levels of organization. - Students will develop critical thinking skills to formulate scientific questions. - Students will communicate results of scientific investigations effectively to scientific and non-scientific audiences in both the oral and written form.			
Course Title	ZOOMJ83: Chronobiology			
Category of Course	Major/ Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- Develop an understanding of the history and scope of chronobiology, including geophysical and biological rhythms. - Study the structure and function of circadian timing systems, focusing on central and peripheral clocks, the roles of sensory receptors, pacemakers, and melatonin. - Evaluate circadian rhythms' impact on homeostasis, disruptions, seasonal affective disorders, and the importance of chronopharmacology, chronomedicine, and chronotherapy.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Introduction to Chronobiology 1.1 History, significant milestones, and scope of the science of chronobiology 1.2 Geophysical and environmental rhythms 1.3 Biological Rhythms: types with examples 1.4 Clock and calendar rhythms			6
II	2. General Features and Characteristics of Rhythms 2.1 Frequency, period, amplitude and phase 2.2 Free running rhythms, phase shifts, phase response curves and phase transition curves 2.3 Photic and non-photic <i>Zeitgebers</i> 2.4 Entrainment: parameter and non-parametric, photic and non-photic mechanisms of photic and non-photic entrainment			8
III	3. Photoperiodism and seasonality 3.1 Ultimate and proximate factors 3.2 Type (I &II) seasonal rhythms 3.3 Models of photoperiodic time measurement and photoperiodic history 3.4 Photoperiodic and seasonal responses in animals			4
IV	4. Structural and functional organization of the circadian timing system 4.1 Central and peripheral clocks 4.2 Structural and functional anatomy of the suprachiasmatic nucleus (SCN) 4.3 Sensory receptors, pacemakers, afferent and efferent pathways 4.4 Melatonin: a clock input and clock output 4.5 Molecular transcriptional-translational-post-translational feedback loops in endogenous time generation (fruit flies, birds, and mammals as examples)			10
V	5. Chronobiology in health and disease 5.1 Homeostasis and circadian rhythmicity (regulation of body temperature, sleep, and energy expenditure) 5.2 Circadian disruption (Jet lag and shift work) and lifestyle disorders 5.3 Seasonal affective disorders and depression 5.4 Concept of chrono-pharmacology, chrono-medicine, and chronotherapy 5.5 Light pollution, circadian clock, and biodiversity			8
Texts / References	1. Circadian Physiology: Roberto Refinetti, CRC Press (3 rd Ed.) 2016. 2. Chronobiology: Biological Timekeeping; Jay C. Dunlap <i>et al.</i> , SinauerAsso.,			

	Inc. Publishers, USA, (2005).			
	3. Introducing Biological Rhythms: Willard L. Koukkari and Robert B. Sothorn, Springer (2006).			
Learning Outcomes (not more than 100 words)	1. Comprehend the history, milestones, and scope of chronobiology, and distinguish between geophysical, environmental, and biological rhythms. 2. Identify the structure and function of circadian timing systems, including central and peripheral clocks and the role of the SCN and melatonin. 3. Assess the impact of circadian rhythms on health, including the effects of circadian disruption, seasonal affective disorders, and applications in chrono-pharmacology and chronotherapy.			
Course Title	ZOOMJ84: Entomology and Fish Biology			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above):			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		4
	Hour of Teaching (Total)	45		75
Course Objectives (not more than 100 words)	The purpose of the course is: - To study the diversity and unique morphological features of different insect orders. - To understand the need of various physiological systems in insects. - To learn about the importance of useful and harmful insects. - To give broad introduction to fish distribution, anatomy and physiology. - Familiarization with fish adaptations and their reproductive strategies.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1.1. Importance and taxonomic richness of insects 1.2. External anatomy 1.2.1. Segmentation and tagmosis 1.2.2. Integument: structure and functions of cuticles, sclerotization			05
II	2. Internal anatomy and physiology 2.1 Nervous system 2.2 Endocrine system and functions of hormones 2.3 Circulatory system: heart and haemolymph 2.4 Respiratory system 2.4.1 Aerial respiration 2.4.2 Aquatic respiration 2.5 Digestive system 2.5.1 Structure of gut 2.5.2 Digestion of food 2.6 Excretory system and waste disposal 2.6.1 Malpighian tubules 2.6.2 Nitrogen excretion 2.7 Reproductive system 2.7.1 Female and male reproductive systems 2.8. Spermatogenesis and oogenesis			07
III	3.1. Sensory system 3.1.1. Tactile mechanoreceptors and position receptors 3.1.2. Compound eye 3.2. Sound and light production in insects 3.3. Insects as friends and foes 3.4. General methods of insect pest management			08
IV	4.1. Natural history and diversity of fishes 4.2. Exotic, larvivorous and weed fishes 4.3. Venomous and poisonous fishes 4.4. Hill stream and Deep-Sea adaptations in fishes 4.5. Migration in fishes			08
V	5.1. Integument – Chromatophores, pigments and scales 5.2. Organs of gas exchange			11

	5.3. Electric organs 5.4. Hydrostatic organs 5.5. Reproductive strategies & Parental care			
Texts / References	<i>Entomology</i> 1. Richards & Davies, Imms' General Textbook of Entomology (Vol. I & II), Chapman & Hall, UK 2. Gullan & Cranston, The Insects: An outline of Entomology, Wiley Blackwell, USA 3. Chapman, The Insects: Structure and Function, Cambridge University Press, UK <i>Fish Biology</i> 1. Lagler, Bardach, Miller and May Passino: Ichthyology (2003, Wiley) 2. Khanna and Singh: A textbook of Fish Biology and Fisheries (2015, Narendra Publishing House) 3. Fish and Fisheries: Pandey and Shukla (Rastogi Publication) 4. Pillay: Aquaculture: Principles and Practices Fishing News Books (2005, First Indian reprint) 5. Srivastava: A Textbook of Fishery Science and Indian Fisheries (1985, Kitab Mahal) 6. Parihar: Fish Biology and Indian fisheries (1999, Central publishing House Allahabad)			
Learning Outcomes (not more than 100 words)	After studying the course, a student will: 1. Learn the vast diversity of insects and their role in environmental engineering. 2. Know the various physiological systems operating in insects. 3. Understand different kinds of beneficial insects, and the measures to control the harmful insects. 4. Become aware of the major research topics in pure and applied entomology. 5. Students will learn about the history, biogeography, physiology and diversity of fish, as well established model system for research.			
Course Title	ZOOMJ85: Lab. exercises based on course ZOOMJ81			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	- To learn basics of instrumentation including principle and applications in the field of Biological Sciences. - Familiarization with cell culture techniques and methodology.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Direct spectrophotometric detection of proteins and nucleic acids Verification of laws of spectrophotometry Separation of proteins using SDS-PAGE			10
II	Demonstration of DNA gel electrophoresis Demonstration of bright field and phase contrast Demonstration of fluorescence, confocal and electron microscope			10
III	Tissue fixation, paraffin block preparation and sectioning Staining of paraffin section using hematoxylin and eosin stain Visit to any tissue culture lab			10
Texts / References	1. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,) 2. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-Livingstones) 3. Bancroft: Theory and Practice of Histological Techniques (2018 Kim S. Suvarna, Christopher Layto, John D. Bancroft) 4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill) 5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)			
Learning Outcomes (not more than 100 words)	Students will be equipped to handle various instruments and learn techniques related to sampling, imaging, data collection, analysis and interpretation in biological samples			
Course Title	ZOOMJ86: Lab. exercises based on course ZOOMJ82 & ZOOMJ83			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			

Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits-Part A		2	2
	Hour of Teaching (Total)		60	60
Course Objectives (not more than 100 words)	The course aims to introduce students to the scientific concepts underlying the study of neural phenomena; provide an understanding of the principles underlying cellular, molecular, and cognitive neurobiology			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	<i>PART-A: Neurobiology</i> - Studies on action potential and measurement of nerve impulse conduction. - Studies on spatial learning and memory in rat/mouse model by Morris-Water/Radial-Arm Maze Tests. - Evaluation of mini-mental status examination. a) External morphology of rat brain b) Staining of neuronal cells - Spectrophotometric assay of acetylcholine esterase. - Demonstration of Primary culture of neurons			30
II	<i>PART-B: Chronobiology</i> - Measurement of circadian locomotor activity rhythm in insects (demonstration). - Measurement of circadian locomotor activity rhythm in mice (demonstration). - Measurement of Human circadian rhythm using Actiwatch/Wristband. - Calculation of period length, phase shifts, and phase angle difference on pre-recorded data. - Construction of a Phase response curve using pre-recorded circadian activity rhythm data. - Observation of circadian rhythms in crickets. - Study of photoperiodic effects on anxiety using Open Field Test/Elevated Plus maze test - Study of re-entrainment of circadian activity rhythms of animals on pre-recorded data.			30
Texts / References	Mark F. Bear (Author), Barry W. Connors (Author), Michael A. Paradiso (Author) Neuroscience: Exploring the Brain (4 th Edition 2015 Wolters Kluwer)			
Learning Outcomes (not more than 100 words)	Students will engage in laboratory investigations that focus on neuroscience. Students will understand how to construct testable hypotheses and design scientific investigations that contribute to neurobiology.			

Course Title	ZOOMJ87: Lab. exercises based on course ZOOMJ84			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above):			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	The purpose of the course is: • To explore the variety and distinctive physical characteristics of various insect orders. • To comprehend the necessity of different physiological systems in insects. • To give broad introduction to fish distribution, anatomy and physiology. • Familiarization with fish adaptations and their reproductive strategies.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1.1. Collection and preservation of insects 1.2. Study of the external morphology of insects			01
II	2.1. Internal anatomy of insects 2.1.1. Alimentary canal 2.1.2. Salivary apparatus: dissection and <i>in-toto</i> preparation 2.2. Dissection of frontal ganglion, brain, retrocerebral endocrine glands and recurrent nerve 2.3. Dissection of prothoracic gland 2.4. Dissection of male and female reproductive systems			03
III	3. Study of external morphology and dissection of sting apparatus of honey bee			01
IV	4.1. Classification of the following locally available fishes using key 4.1.1. Carps: <i>Catla catla</i> ; <i>Labeo rohita</i> ; <i>Cirrhina mrigala</i> 4.1.2. Catfishes: <i>Heteropneustes fossilis</i> ; <i>Clarias batrachus</i> 4.2. Study of museum specimens of fishes having electric organs, venomous organs and air breathing organs			04
V	5.1. Dissection and display of accessory respiratory organs of 5.1.1. <i>Clarias batrachus</i> 5.1.2. <i>Heteropneustes fossilis</i> 5.2. Mounting of respiratory epithelium of accessory respiratory organs of <i>H. fossilis</i> 5.3. Study of T.S. of gills, accessory respiratory organs and swim bladder from prepared slides			04
Texts / References	<i>Entomology</i> 1. Richards & Davies, Imms' General Textbook of Entomology (Vol. I & II), Chapman & Hall, UK 2. Gullan & Cranston, The Insects: An outline of Entomology, Wiley Blackwell, USA 3. Chapman, The Insects: Structure and Function, Cambridge University Press, UK <i>Fish Biology</i> 1. Lagler, Bardach, Miller and May Passino: Ichthyology (2003, Wiley) 2. Khanna and Singh: A textbook of Fish Biology and Fisheries (2015, Narendra Publishing House) 3. Fish and Fisheries: Pandey and Shukla (Rastogi Publication) 4. Pillay: Aquaculture: Principles and Practices Fishing News Books (2005, First Indian reprint) 5. Srivastava: A Textbook of Fishery Science and Indian Fisheries (1985, Kitab Mahal) 6. Parihar: Fish Biology and Indian fisheries (1999, Central publishing House Allahabad)			
Learning Outcomes (not more than 100 words)	After studying the course, a student will: 1. Gain knowledge of the different physiological systems present in insects. 2. Learn about various beneficial insects and familiarize with the key research areas in practical entomology. 3. Students will learn about the history, biogeography, physiology and diversity of fish, a well-established model system for research.			

Course Title	ZOOMN81: Biotechniques			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		3
	Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)	- To learn basics of instrumentation including principle and applications in the field of Biological Sciences. - Familiarization with cell culture techniques and methodology.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	1. Analytical instruments: Principle and applications& Microscopy 1.1 pH meter 1.2 UV-visible spectrophotometer 1.3 Centrifuge 1.4 Brightfield 1.5 Darkfield 1.6 Polarized and differential interference contrast microscopy 1.7 Atomic force microscopy 1.8 Fluorescence and Confocal 1.9 Scanning and transmission electron microscopes			15
II	2. Histological Techniques 2.1 Fixation and Fixatives 2.2 Tissue processing and Microtomy 2.3 Principle and methods of histological staining 2.3.1 The Haematoxylin and eosin 2.4 Principles of immunohistochemistry			7
III	3. Cell and tissue culture techniques 3.1 Sterilization: laminar flow, media and glassware 3.2 Types of animal cell culture 3.3 Cell viability testing 3.4 Cryopreservation			8
IV	4. Separation techniques 4.1 Chromatography: Principle and Applications 4.1.1 Paper chromatography 4.1.2 Thin layer chromatography 4.1.3 Ion exchange chromatography 4.1.4 Gel-filtration chromatography 4.1.5 HPLC 4.2 Electrophoresis: Principle and applications of 4.2.1 Agarose gel electrophoresis, native and SDS-PAGE 4.2.2 Western blotting			10
V	5. Radiotracer and Immunoassay techniques: Principle and Applications 5.1 Radioactivity: Nature, measurements and safety aspects 5.2 Immunoassay techniques 5.2.1 RIA 5.2.2 ELISA			5
Texts / References	1. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,) 2. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-Livingstones) 3. Bancroft: Theory and Practice of Histological Techniques (2018 Kim S. Suvarna, Christopher Layto, John D. Bancroft) 4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill) 5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)			
Learning Outcomes (not more than 100 words)	Students will be equipped to handle various instruments and learn techniques related to sampling, imaging, data collection, analysis and interpretation in biological samples			

Course Title	ZOOMN82: Lab. exercises based on course ZOOMN81			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	- To learn basics of instrumentation including principle and applications in the field of Biological Sciences. - Familiarization with cell culture techniques and methodology.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Direct spectrophotometric detection of proteins and nucleic acids Verification of laws of spectrophotometry Separation of proteins using SDS-PAGE			10
II	Demonstration of DNA gel electrophoresis Demonstration of bright field and phase contrast Demonstration of fluorescence, confocal and electron microscope			10
III	Tissue fixation, paraffin block preparation and sectioning Staining of paraffin section using hematoxylin and eosin stain Visit to any tissue culture lab			10
Texts / References	1. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,) 2. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-Livingstones) 3. Bancroft: Theory and Practice of Histological Techniques (2018 Kim S. Suvarna, Christopher Layto, John D. Bancroft) 4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill) 5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)			
Learning Outcomes (not more than 100 words)	Students will be equipped to handle various instruments and learn techniques related to sampling, imaging, data collection, analysis and interpretation in biological samples			

Course Title		ZOOMJ8R1: Biotechniques			
Category of Course		Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching			Theory	Practical	Cumulative
		Credits	3		3
		Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)		- To learn basics of instrumentation including principle and applications in the field of Biological Sciences. - Familiarization with cell culture techniques and methodology.			
Course Content		The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content				Hr. of Teaching
I	1. Analytical instruments: Principle and applications& Microscopy 1.1 pH meter 1.2 UV-visible spectrophotometer 1.3 Centrifuge 1.4 Brightfield 1.5 Darkfield 1.6 Polarized and differential interference contrast microscopy 1.7 Atomic force microscopy 1.8 Fluorescence and Confocal 1.9 Scanning and transmission electron microscopes				15
II	2. Histological Techniques 2.1 Fixation and Fixatives 2.2 Tissue processing and Microtomy 2.3 Principle and methods of histological staining 2.3.1 The Haematoxylins and eosin 2.4 Principles of immunohistochemistry				7
III	3. Cell and tissue culture techniques 3.1 Sterilization: laminar flow, media and glassware 3.2 Types of animal cell culture 3.3 Cell viability testing 3.4 Cryopreservation				8
IV	4. Separation techniques 4.1 Chromatography: Principle and Applications 4.1.1 Paper chromatography 4.1.2 Thin layer chromatography 4.1.3 Ion exchange chromatography 4.1.4 Gel-filtration chromatography 4.1.5 HPLC 4.2 Electrophoresis: Principle and applications of 4.2.1 Agarose gel electrophoresis, native and SDS-PAGE 4.2.2 Western blotting				10
V	5. Radiotracer and Immunoassay techniques: Principle and Applications 5.1 Radioactivity: Nature, measurements and safety aspects 5.2 Immunoassay techniques 5.2.1 RIA 5.2.2 ELISA				5
Texts / References		1. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,) 2. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-Livingstones) 3. Bancroft: Theory and Practice of Histological Techniques (2018 Kim S. Suvarna, Christopher Layto, John D. Bancroft) 4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill) 5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)			
Learning Outcomes (not more than 100 words)		Students will be equipped to handle various instruments and learn techniques related to sampling, imaging, data collection, analysis and interpretation in biological samples			

Course Title	ZOOMJ8R2: Lab exercises based on course ZOOMJ8R1			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	- To learn basics of instrumentation including principle and applications in the field of Biological Sciences. - Familiarization with cell culture techniques and methodology.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Direct spectrophotometric detection of proteins and nucleic acids Verification of laws of spectrophotometry Separation of proteins using SDS-PAGE			10
II	Demonstration of DNA gel electrophoresis Demonstration of bright field and phase contrast Demonstration of fluorescence, confocal and electron microscope			10
III	Tissue fixation, paraffin block preparation and sectioning Staining of paraffin section using hematoxylin and eosin stain Visit to any tissue culture lab			10
Texts / References	1. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,) 2. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-Livingstones) 3. Bancroft: Theory and Practice of Histological Techniques (2018 Kim S. Suvarna, Christopher Layto, John D. Bancroft) 4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill) 5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)			
Learning Outcomes (not more than 100 words)	Students will be equipped to handle various instruments and learn techniques related to sampling, imaging, data collection, analysis and interpretation in biological samples			

Course Title		ZOOMN8R1: Biotechniques			
Category of Course		Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching			Theory	Practical	Cumulative
		Credits	3		3
		Hour of Teaching (Total)	45		45
Course Objectives (not more than 100 words)		- To learn basics of instrumentation including principle and applications in the field of Biological Sciences. - Familiarization with cell culture techniques and methodology.			
Course Content		The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content				Hr. of Teaching
I	1. Analytical instruments: Principle and applications& Microscopy 1.1 pH meter 1.2 UV-visible spectrophotometer 1.3 Centrifuge 1.4 Brightfield 1.5 Darkfield 1.6 Polarized and differential interference contrast microscopy 1.7 Atomic force microscopy 1.8 Fluorescence and Confocal 1.9 Scanning and transmission electron microscopes				15
II	2. Histological Techniques 2.1 Fixation and Fixatives 2.2 Tissue processing and Microtomy 2.3 Principle and methods of histological staining 2.3.1 The Haematoxylins and eosin 2.4 Principles of immunohistochemistry				7
III	3. Cell and tissue culture techniques 3.1 Sterilization: laminar flow, media and glassware 3.2 Types of animal cell culture 3.3 Cell viability testing 3.4 Cryopreservation				8
IV	4. Separation techniques 4.1 Chromatography: Principle and Applications 4.1.1 Paper chromatography 4.1.2 Thin layer chromatography 4.1.3 Ion exchange chromatography 4.1.4 Gel-filtration chromatography 4.1.5 HPLC 4.2 Electrophoresis: Principle and applications of 4.2.1 Agarose gel electrophoresis, native and SDS-PAGE 4.2.2 Western blotting				10
V	5. Radiotracer and Immunoassay techniques: Principle and Applications 5.1 Radioactivity: Nature, measurements and safety aspects 5.2 Immunoassay techniques 5.2.1 RIA 5.2.2 ELISA				5
Texts / References		1. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,) 2. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-Livingstones) 3. Bancroft: Theory and Practice of Histological Techniques (2018 Kim S. Suvarna, Christopher Layto, John D. Bancroft) 4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill) 5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)			
Learning Outcomes (not more than 100 words)		Students will be equipped to handle various instruments and learn techniques related to sampling, imaging, data collection, analysis and interpretation in biological samples			

Course Title	ZOOMN8R2: Lab. exercises based on course ZOOMN8R1			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	1
	Hour of Teaching (Total)		30	30
Course Objectives (not more than 100 words)	- To learn basics of instrumentation including principle and applications in the field of Biological Sciences. - Familiarization with cell culture techniques and methodology.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Direct spectrophotometric detection of proteins and nucleic acids Verification of laws of spectrophotometry Separation of proteins using SDS-PAGE			10
II	Demonstration of DNA gel electrophoresis Demonstration of bright field and phase contrast Demonstration of fluorescence, confocal and electron microscope			10
III	Tissue fixation, paraffin block preparation and sectioning Staining of paraffin section using hematoxylin and eosin stain Visit to any tissue culture lab			10
Texts / References	1. Boyer: Modern Experimental Biochemistry (1993, Benjamin-Cummings,) 2. Pearse: Histochemistry - Theoretical and applied, Volume I-III (1980-1993, Churchill-Livingstones) 3. Bancroft: Theory and Practice of Histological Techniques (2018 Kim S. Suvarna, Christopher Layto, John D. Bancroft) 4. Plummer: An Introduction to Practical Biochemistry (1989, McGraw Hill) 5. Wilson & Walker: Experimental Biochemistry (2006, Cambridge)			
Learning Outcomes (not more than 100 words)	Students will be equipped to handle various instruments and learn techniques related to sampling, imaging, data collection, analysis and interpretation in biological samples			

Course Title	ZOODS8R3: Dissertation			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		12	12
	Hour of Teaching (Total)		360	360
Course Objectives (not more than 100 words)	- To provide opportunity to utilize their learned skills in the class room into practice and to enhance their skills in the research environment/field work and benefit from the Mentor/Supervisor expertise and developing a learning network. - To provide opportunity to go through the literature/field observation, set up the experiments, obtaining the results and analyzing and interpretation of results and putting together in a scientific write-up. - To provide opportunity to present their experimental/field observation results before the examiners.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
	Units	Course Content	Hr. of Teaching	
	I	1. Research projects under the guidance of a faculty member. 2. The student has to select a topic for the dissertation from the Core Course (CC). The dissertation includes laboratory/field-based work followed by the submission of the report and presentation.	24 hr./week	
Texts / References				
Learning Outcomes (not more than 100 words)	Students will be learnt to go through the literature/field observations, asking questions, framing objectives, using appropriate methods to carry out experiments, obtaining results, analyzing, discussing the results and preparing their own presentation.			